

Q1 The definitions in an XML document are said to be _____ when the tagging system and definitions in the DTD are all in compliance.

- (A) well-formed
- (B) reasonable
- (C) valid
- (D) logical

Q2 Consider the JavaScript Code :

```
var y="12";
function f() {
    var y='6';
    alert (this.y);
    function g() {alert (y); }
    g();
}
f();
```

If M is the number of alert dialog boxes generated by this JavaScript code and $D_1, D_2, \dots, D_M$ represents the content displayed in each of the M dialog boxes, then :

- (A) $M=3$; D_1 displays "12"; D_2 displays "6"; D_3 displays "12".
- (B) $M=3$; D_1 displays "6"; D_2 displays "12"; D_3 displays "6".
- (C) $M=2$; D_1 displays "6"; D_2 displays "12".
- (D) $M=2$; D_1 displays "12"; D_2 displays "6"

Q3 What is the output of the following JAVA program ?

```
class simple
{
    public static void main(String[] args)
    {
        simple obj = new simple();
        obj.start();
    }
    void start()
    {
        long [] P= {3, 4, 5};
        long [] Q= method (P);
        System.out.print (P[0] + P[1] + P[2]+":");
```

```
System.out.print (Q[0] + Q[1] + Q[2]);
    }
    long [] method (long [] R)
    {
        R [1]=7;
        return R;
    }
} //end of class
```

- (A) 12 : 15
- (B) 15 : 12
- (C) 12 : 12
- (D) 15 : 15

Q4 What is the output of the following 'C' program ? (Assuming little - endian representation of multi-byte data in which Least Significant Byte (LSB) is stored at the lowest memory address.)

```
#include <stdio.h>
#include <stdlib.h>
/* Assume short int occupies two bytes of storage */
int main ()
{
    union saving
    {
        short int one;
        char two[2];
    };
    union saving m;
    m.two [0] = 5;
    m.two [1] = 2;
    printf("%%d, %%d, %%d\n", m.two [0], m.two [1], m.one);
} /* end of main */
```

- (A) 5, 2, 1282
- (B) 5, 2, 52
- (C) 5, 2, 25
- (D) 5, 2, 517

Q5 Given below are three implementations of the swap() function in C++ :

(a)	(b)	(c)
void swap (int a, int b)	void swap (int &a, int &b)	void swap (int *a, int *b)
{ int temp;	{	{



temp = a; a = b; b = temp; }	int temp; temp = a; a = b; b = temp; }	int *temp; temp = a; a = b; b = temp; }
int main() { int p = 0, q = 1; swap (p, q); }	int main() { int p = 0, q = 1; swap (p, q); }	int main() { int p = 0, q = 1; swap (&p, &q); }

Which of these would actually swap the contents of the two integer variables p and q ?

- (A) (a) only
(B) (b) only
(C) (c) only
(D) (b) and (c) only

Q6 In Java, which of the following statements is/are True ?

S1 : The 'final' keyword applied to a class definition prevents the class from being extended through derivation.
S2 : A class can only inherit one class but can implement multiple interfaces.
S3 : Java permits a class to replace the implementation of a method that it has inherited. It is called method overloading.

- (A) S1 and S2 only
(B) S1 and S3 only
(C) S2 and S3 only
(D) All of S1, S2 and S3

Q7 Which of the following statements is/are True ?

P : C programming language has a weak type system with static types.

Q : Java programming language has a strong type system with static types.

- (A) P only
(B) Q only
(C) Both P and Q
(D) Neither P nor Q

Q8 A graphic display system has a frame buffer that is 640 pixels wide, 480 pixels high and 1 bit of color depth. If the access time for each pixel on the average is 200 nanoseconds, then the refresh rate of this frame buffer is approximately :

- (A) 16 frames per second
(B) 19 frames per second
(C) 21 frames per second
(D) 23 frames per second

Q9 Which of the following statements is/are True regarding the solution to the visibility problem in 3D graphics ?

S1 : The Painter's algorithm sorts polygons by depth and then paints (scan - converts) each Polygon on to the screen starting with the most nearest polygon.

S2 : Backface Culling refers to eliminating geometry with backfacing normals.

- (A) S1 only
(B) S2 only
(C) Both S1 and S2
(D) Neither S1 nor S2

Q10

$$\text{Consider the matrix } M = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

representing a set of planar (2D) geometric transformations in homogeneous coordinates. Which of the following statements about the matrix M is True ?

- (A) M represents first, a scaling of vector (2, 1) followed by translation of vector (1, 1)
(B) M represents first, a translation of vector (1, 1) followed by scaling of vector (2, 1)
(C) M represents first, a scaling of vector (3, 1) followed by shearing of parameters (-1, 1)
(D) M represents first, a shearing of parameters (-1, 1) followed by scaling of vector (3, 1)

Q11 Assume the following regarding the development of a software system P :

- Estimated lines of code of P : 33, 480 LOC
- Average productivity for P : 620 LOC per



person-month

- Number of software developers : 6

- Average salary of a software developer : ₹ 50,000 per month

If E, D and C are the estimated development effort (in person-months), estimated development time (in months), and estimated development cost (in ₹`Lac) respectively, then (E, D, C) = _____.

- (A) (48, 8, 24)
(B) (54, 9, 27)
(C) (60, 10, 30)
(D) (42, 7, 21)

Q12 Match the following in Software Engineering :

	List I		List II
(a)	Product Complexity	(i)	Software Requirement Definition
(b)	Structured System Analysis	(ii)	Software Design
(c)	Coupling and Cohesion	(iii)	Validation Technique
(d)	Symbolic Execution	(iv)	Software Cost Estimation

- (A) (a) (b) (c) (d)
(ii) (iii) (iv) (i)
(B) (a) (b) (c) (d)
(iii) (i) (iv) (ii)
(C) (a) (b) (c) (d)
(iv) (i) (ii) (iii)
(D) (a) (b) (c) (d)
(iii) (iv) (i) (ii)

Q13 Which one of the following is not typically provided by Source Code Management Software ?

- (A) Synchronisation
(B) Versioning and Revision history
(C) Syntax highlighting
(D) Project forking

Q14 A software system crashed 20 times in the year 2017 and for each crash, it took 2 minutes to

restart. Approximately, what was the software availability in that year ?

- (A) 96.9924% (B) 97.9924%
(C) 98.9924% (D) 99.9924%

Q15 Match the 5 CMM Maturity levels/CMMI staged representations in List- I with their characterizations in List-II

	List I		List II
(a)	Initial	(i)	Processes are improved quantitatively and continually.
(b)	Repeatable	(ii)	The plan for a project comes from a template for plans.
(c)	Defined	(iii)	The plan uses processes that can be measured quantitatively.
(d)	Managed	(iv)	There may not exist a plan or it may be abandoned.
(e)	Optimizing	(v)	There's a plan and people stick to it.

- (A) (a) (b) (c) (d) (e)
(iv) (v) (i) (iii) (ii)
(B) (a) (b) (c) (d) (e)
(i) (ii) (iv) (v) (iii)
(C) (a) (b) (c) (d) (e)
(v) (iv) (ii) (iii) (i)
(D) (a) (b) (c) (d) (e)
(iv) (v) (ii) (iii) (i)

Q16 Coupling is a measure of the strength of the interconnections between software modules. Which of the following are correct statements with respect to module coupling ?

P : Common coupling occurs when one module controls the flow of another module by passing it information on what to do.

Q : In data coupling, the complete data structure is passed from one module to another through parameters.

R : Stamp coupling occurs when modules share a composite data structure and use only parts of it.



- (A) P and Q only
(B) P and R only
(C) Q and R only
(D) All of P, Q and R
- Q17** A software design pattern often used to restrict access to an object is :
(A) adapter
(B) decorator
(C) delegation
(D) proxy
- Q18** Reasons to re-engineer a software include :
P : Allow legacy software to quickly adapt to the changing requirements
Q : Upgrade to newer technologies/platforms/paradigm (for example, object-oriented)
R : Improve software maintainability
S : Allow change in the functionality and architecture of the software
(A) P, R and S only
(B) P and R only
(C) P, Q and S only
(D) P, Q and R only
- Q19** Which of the following is not a key strategy followed by the clean room approach to software development ?
(A) Formal specification
(B) Dynamic verification
(C) Incremental development
(D) Statistical testing of the system
- Q20** Which of the following statements is/are True ?
P : Refactoring is the process of changing a software system in such a way that it does not alter the external behavior of the code yet improves the internal architecture.
Q : An example of refactoring is adding new features to satisfy a customer requirement discovered after a project is shipped.
(A) P only
(B) Q only
(C) Both P and Q
(D) Neither P nor Q
- Q21** The solution of the recurrence relation $T(m) = T(3m/4) + 1$ is :
(A) $\theta(\lg m)$
(B) $\theta(m)$
(C) $\theta(\lg m)$
(D) $\theta(\lg \lg m)$
- Q22** Consider the array $A = \langle 4, 1, 3, 2, 16, 9, 10, 14, 8, 7 \rangle$. After building heap from the array A, the depth of the heap and the right child of max-heap are _____ and _____ respectively. (Root is at level 0).
(A) 3, 14
(B) 3, 10
(C) 4, 14
(D) 4, 10
- Q23** A hash function h defined $h(\text{key}) = \text{key} \bmod 7$, with linear probing, is used to insert the keys 44, 45, 79, 55, 91, 18, 63 into a table indexed from 0 to 6. What will be the location of key 18 ?
(A) 3
(B) 4
(C) 5
(D) 6
- Q24** Which of the following algorithms solves the single-source shortest paths ?
(A) Prim's algorithm
(B) Floyd - Warshall algorithm
(C) Johnson's algorithm
(D) Dijkstra's algorithm
- Q25** A text is made up of the characters A, B, C, D, E each occurring with the probability 0.08, 0.40, 0.25, 0.15 and 0.12 respectively. The optimal coding technique will have the average length of :
(A) 2.4
(B) 1.87
(C) 3.0
(D) 2.15
- Q26** A binary search tree in which every non-leaf node has non-empty left and right subtrees is called a strictly binary tree. Such a tree with 19 leaves :
(A) cannot have more than 37 nodes
(B) has exactly 37 nodes
(C) has exactly 35 nodes
(D) cannot have more than 35 nodes
- Q27**



Match the following with respect to algorithm paradigms :

List - I	List - II
(a) The 8-Queen's problem	(i) Dynamic Programming
(b) Single-Source shortest paths	(ii) Divide and conquer
(c) STRASSEN's Matrix multiplication	(iii) Greedy approach
(d) Optimal binary search trees	(iv) Backtracking

(A) (a) (b) (c) (d)
(iv) (i) (iii) (ii)

(B) (a) (b) (c) (d)
(iv) (iii) (i) (ii)

(C) (a) (b) (c) (d)
(iii) (iv) (ii) (i)

(D) (a) (b) (c) (d)
(iv) (iii) (ii) (i)

Q28 The maximum number of comparisons needed to sort 9 items using radix sort is (assume each item is 5 digit octal number) :

- (A) 45 (B) 72
(C) 360 (D) 450

Q29 A 5-ary tree is tree in which every internal node has exactly 5 children. The number of leaf nodes in such a tree with 8 internal nodes will be :

- (A) 30 (B) 33
(C) 45 (D) 125

Q30 Consider a Boolean function of 'n' variables. The order of an algorithm that determines whether the Boolean function produces a output 1 is :

- (A) Logarithmic
(B) Linear
(C) Quadratic
(D) Exponential

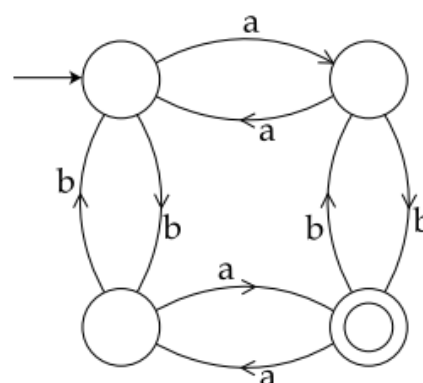
Q31 Two finite state machines are said to be equivalent if they :

- (A) Have the same number of edges
(B) Have the same number of states

(C) Recognize the same set of tokens

(D) Have the same number of states and edges

Q32 The finite state machine given in figure below recognizes :



- (A) any string of odd number of a's
(B) any string of odd number of b's
(C) any string of even number of a's and odd number of b's
(D) any string of odd number of a's and odd number of b's

Q33 A pushdown automata behaves like a Turing machine when the number of auxiliary memory is :

- (A) 0 (B) 1
(C) 1 or more (D) 2 or more

Q34 Pushdown automata can recognize language generated by_____.

- (A) Only context free grammar
(B) Only regular grammar
(C) Context free grammar or regular grammar
(D) Only context sensitive grammar

Q35 To obtain a string of n Terminals from a given Chomsky normal form grammar, the number of productions to be used is :

- (A) $2n-1$ (B) $2n$
(C) $n+1$ (D) n^2

Q36 Consider the following two Grammars :

$G_1 : S \rightarrow SbS \mid a$

$G_2 : S \rightarrow aB \mid ab, A \rightarrow GAB \mid a, B \rightarrow ABb \mid b$

Which of the following option is correct ?



- (A) Only G_1 is ambiguous
 (B) Only G_2 is ambiguous
 (C) Both G_1 and G_2 are ambiguous
 (D) Both G_1 and G_2 are not ambiguous
- Q37** Context-sensitive language can be recognized by a :
 (A) Finite state machine
 (B) Deterministic finite automata
 (C) Non-deterministic finite automata
 (D) Linear bounded automata
- Q38** The set $A = \{ 0^n 1^n 2^n \mid n = 1, 2, 3, \dots \}$ is an example of a grammar that is :
 (A) Context-sensitive
 (B) Context-free
 (C) Regular
 (D) None of the above
- Q39** A bottom-up parser generates :
 (A) Left-most derivation in reverse
 (B) Right-most derivation in reverse
 (C) Left-most derivation
 (D) Right-most derivation
- Q40** Consider the following statements () :
 S_1 : There exists no algorithm for deciding if any two Turing machines M_1 and M_2 accept the same language.
 S_2 : The problem of determining whether a Turing machine halts on any input is undecidable.
 Which of the following options is correct?
 (A) Both S_1 and S_2 are correct
 (B) Both S_1 and S_2 are not correct
 (C) Only S_1 is correct
 (D) Only S_2 is correct
- Q41** A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200 Kbps bandwidth. Find the throughput of the system, if the system (all stations put together) produces 250 frames per second :
 (A) 49 (B) 368
 (C) 149 (D) 151
- Q42** The period of a signal is 100 ms. Its frequency is _____.

- (A) 100^3 Hertz
 (B) 10^{-2} KHZ
 (C) 10^{-3} KHZ
 (D) 10^5 Hertz

- Q43** The dotted-decimal notation of the following IPV4 address in binary notation is _____.
 10000001 00001011 00001011 11101111
 (A) 111.56.45.239
 (B) 129.11.10.238
 (C) 129.11.11.239
 (D) 111.56.11.239
- Q44** Which of the following statements are true ?
 (a) Advanced Mobile Phone System (AMPS) is a second generation cellular phone system.
 (b) IS - 95 is a second generation cellular phone system based on CDMA and DSSS.
 (c) The Third generation cellular phone system will provide universal personnel communication.
 (A) (a) and (b) only
 (B) (b) and (c) only
 (C) (a), (b) and (c)
 (D) (a) and (c) only

- Q45** Match the following symmetric block ciphers with corresponding block and key sizes :

	List - I	List - II
(a)	DES	(i) block size 64 and key size ranges between 32 and 448.
(b)	IDEA	(ii) block size 64 and key size 64.
(c)	BLOWFISH	(iii) block size 128 and key sizes 128, 192, 256.
(d)	AES	(iv) block size 64 and key size 128.

- (A) a- iv, b-ii, c-i, d-iii
 (B) a- ii, b-iv, c-i, d-iii
 (C) a- ii, b-iv, c-iii, d-i
 (D) a- iv, b-ii, c-iii, d-i

Q46



Which of the following statements is/are true?

- (a) Three broad categories of Networks are
 (i) Circuit Switched Networks
 (ii) Packet Switched Networks
 (iii) Message Switched Networks
 (b) Circuit Switched Network resources need not be reserved during the setup phase.
 (c) In packet switching there is no resource allocation for packets.
 (A) (a) and (b) only
 (B) (b) and (c) only
 (C) (a) and (c) only
 (D) (a), (b) and (c)

Q47 In Challenge-Response authentication the claimant _____.

- (A) Proves that she knows the secret without revealing it
 (B) Proves that she doesn't know the secret
 (C) Reveals the secret
 (D) Gives a challenge

Q48 Decrypt the message "WTAAD" using the Caesar Cipher with key=15.

- (A) LIPPS (B) HELLO
 (C) OLLEH (D) DAATW

Q49 To guarantee correction of up to t errors, the minimum Hamming distance $d_{\min}$ in a block code must be _____.

- (A) $t+1$
 (B) $t-2$
 (C) $2t-1$
 (D) $2t+1$

Q50 Encrypt the Message "HELLO MY DEARZ" using a Transposition Cipher with

$$\text{Key} \left\{ \begin{array}{cc} \text{Plain Text} & 2413 \\ \text{Cipher Text} & 1234 \end{array} \right\}$$

- (A) HLLEO YM AEDRZ
 (B) EHOLL ZYM RAED
 (C) ELHL MDOY AZER
 (D) ELHL DOMY ZAER

Q51 At a particular time of computation, the value of a counting semaphore is 10. Then 12 P operations

and "x" V operations were performed on this semaphore. If the final value of semaphore is 7, x will be :

- (A) 8 (B) 9
 (C) 10 (D) 11

Q52 In a paged memory, the page hit ratio is 0.40. The time required to access a page in secondary memory is equal to 120 ns. The time required to access a page in primary memory is 15 ns. The average time required to access a page is _____.

- (A) 105 (B) 68
 (C) 75 (D) 78

Q53 In a multi-user operating system, 30 requests are made to use a particular resource per hour, on an average. The probability that no requests are made in 40 minutes, when arrival pattern is a poisson distribution, is _____.

- (A) e^{-15} (B) $1 - e^{-15}$
 (C) $1 - e^{-20}$ (D) e^{-20}

Q54 Normally user programs are prevented from handling I/O directly by I/O instructions in them. For CPUs having explicit I/O instructions, such I/O protection is ensured by having the I/O instructions privileged. In a CPU with memory mapped I/O, there is no explicit I/O instruction. Which one of the following is true for a CPU with memory mapped I/O ?

- (A) I/O protection is ensured by operating system routines.
 (B) I/O protection is ensured by a hardware trap.
 (C) I/O protection is ensured during system configuration.
 (D) I/O protection is not possible.

Q55 Which UNIX/Linux command is used to make all files and sub-directories in the directory "progs" executable by all users ?

- (A) `chmod -R a+x progs`
 (B) `chmod -R 222 progs`
 (C) `chmod -X a+x progs`
 (D) `chmod -X 222 progs`



- Q56** Which of the following statements are true ?
 (a) External Fragmentation exists when there is enough total memory space to satisfy a request but the available space is contiguous.
 (b) Memory Fragmentation can be internal as well as external.
 (c) One solution to external Fragmentation is compaction.
 (A) (a) and (b) only
 (B) (a) and (c) only
 (C) (b) and (c) only
 (D) (a), (b) and (c)

- Q57** Page information in memory is also called as Page Table. The essential contents in each entry of a page table is/are _____.
 (A) Page Access information
 (B) Virtual Page number
 (C) Page Frame number
 (D) Both virtual page number and Page Frame Number

- Q58** Consider a virtual page reference string 1, 2, 3, 2, 4, 2, 5, 2, 3, 4. Suppose LRU page replacement algorithm is implemented with 3 page frames in main memory. Then the number of page faults are _____.
 (A) 5 (B) 7
 (C) 9 (D) 10

- Q59** Consider the following three processes with the arrival time and CPU burst time given in milliseconds :

Process	Arrival Time	Burst Time
P ₁	0	7
P ₂	1	4
P ₃	2	8

The Gantt Chart for preemptive SJF scheduling algorithm is _____.

- (A)

	P ₁		P ₂		P ₃	
--	----------------	--	----------------	--	----------------	--

0 7 13 21
- (B)

	P ₁		P ₂		P ₁		P ₃	
--	----------------	--	----------------	--	----------------	--	----------------	--

0 1 5 11 19
- (C)

	P ₁		P ₂		P ₃	
--	----------------	--	----------------	--	----------------	--

0 7 11 19

- (D)

	P ₂		P ₃		P ₁	
--	----------------	--	----------------	--	----------------	--

0 4 12 19

- Q60** In which of the following scheduling criteria, context switching will never take place?
 (A) ROUND ROBIN
 (B) Preemptive SJF
 (C) Non-preemptive SJF
 (D) Preemptive priority

- Q61** In RDBMS, which type of Join returns all rows that satisfy the join condition ?
 (A) Inner Join (B) Outer Join
 (C) Semi Join (D) Anti Join

- Q62** Consider a relation book (title, price) which contains the titles and prices of different books. Assuming that no two books have the same price, what does the following SQL query list ?
 Select title
 from book as B
 where (select count (*)
 from book as T
 where T.price > B.price) < 7
 (A) Titles of the six most expensive books.
 (B) Title of the sixth most expensive books.
 (C) Titles of the seven most expensive books.
 (D) Title of the seventh most expensive books.

- Q63** In a Hierarchical database, a hashing function is used to locate the _____.
 (A) Collision
 (B) Root
 (C) Foreign Key
 (D) Records

- Q64** Relations produced from E - R Model will always be in _____.
 (A) 1 NF (B) 2 NF
 (C) 3 NF (D) 4 NF

- Q65** Consider the following schedules involving two transactions.



$S_1 : r_1(X); r_1(Y); r_2(X); r_2(Y); w_2(Y); w_1(X)$

$S_2 : r_1(X); r_2(X); r_2(Y); w_2(Y); r_1(Y); w_1(X)$

Which one of the following statements is correct with respect to above?

- (A) Both S_1 and S_2 are conflict serializable.
- (B) Both S_1 and S_2 are not conflict serializable.
- (C) S_1 is conflict serializable and S_2 is not conflict serializable.
- (D) S_1 is not conflict serializable and S_2 is conflict serializable.

Q66 For a database relation $R(a, b, c, d)$ where the domains of a, b, c and d include only atomic values, and only the following functional dependencies and those that can be inferred from them hold :

$a \rightarrow c$

$b \rightarrow d$

The relation is in _____.

- (A) First normal form but not in second normal form
- (B) Second normal form but not in third normal form
- (C) Third normal form
- (D) BCNF

Q67 A many-to-one relationship exists between entity sets r_1 and r_2 . How will it be represented using functional dependencies if $Pk(r)$ denotes the primary key attribute of relation r ?

- (A) $Pk(r_1) \rightarrow Pk(r_2)$
- (B) $Pk(r_2) \rightarrow Pk(r_1)$
- (C) $Pk(r_2) \rightarrow Pk(r_1)$ and $Pk(r_1) \rightarrow Pk(r_2)$
- (D) $Pk(r_2) \rightarrow Pk(r_1)$ or $Pk(r_1) \rightarrow Pk(r_2)$

Q68 Database systems that store each relation in a separate operating system file may use the operating system's authorization scheme, instead of defining a special scheme themselves. In this case, which of the following is false?

- (A) The administrator enjoys more control on the grant option.
- (B)

It is difficult to differentiate among the update, delete and insert authorizations.

- (C) Cannot store more than one relation in a file.
- (D) Operations on the database are speeded up as the authorization procedure is carried out at the operating system level.

Q69 Let $R_1(a, b, c)$ and $R_2(x, y, z)$ be two relations in which a is the foreign key of R_1 that refers to the primary key of R_2 . Consider following four options.

- (a) Insert into R_1
- (b) Insert into R_2
- (c) Delete from R_1
- (d) Delete from R_2

Which of the following is correct about the referential integrity constraint with respect to above?

- (A) Operations (a) and (b) will cause violation.
- (B) Operations (b) and (c) will cause violation.
- (C) Operations (c) and (d) will cause violation.
- (D) Operations (d) and (a) will cause violation.

Q70 Consider a hash table of size seven, with starting index zero, and a hash function $(7x+3) \bmod 4$. Assuming the hash table is initially empty, which of the following is the contents of the table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing ? Here “_” denotes an empty location in the table.

- (A) 3, 10, 1, 8, _, _, _
- (B) 1, 3, 8, 10, _, _, _
- (C) 1, _, 3, _, 8, _, 10
- (D) 3, 10, _, _, 8, _, _

Q71 In Artificial Intelligence (AI), an environment is uncertain if it is _____.

- (A) Not fully observable and not deterministic
- (B) Not fully observable or not deterministic
- (C) Fully observable but not deterministic
- (D) Not fully observable but deterministic

Q72 In Artificial Intelligence (AI), a simple reflex agent selects actions on the basis of _____.

- (A) current percept, completely ignoring rest of the percept history.



- (B) rest of the percept history, completely ignoring current percept.
 (C) both current percept and complete percept history.
 (D) More than one of the above
 (E) None of the above

Q73 In heuristic search algorithms in Artificial Intelligence (AI), if a collection of admissible heuristics $h_1, \dots, h_m$ is available for a problem and none of them dominates any of the others, which should we choose?

- (A) $h(n) = \max \{h_1(n), \dots, h_m(n)\}$
 (B) $h(n) = \min \{h_1(n), \dots, h_m(n)\}$
 (C) $h(n) = \text{avg} \{h_1(n), \dots, h_m(n)\}$
 (D) $h(n) = \text{sum} \{h_1(n), \dots, h_m(n)\}$

Q74 Consider following sentences regarding A^* , an informed search strategy in Artificial Intelligence (AI).

- (a) A^* expands all nodes with $f(n) < C^*$.
 (b) A^* expands no nodes with $f(n) / C^*$.
 (c) Pruning is integral to A^* .

Here, C^* is the cost of the optimal solution path. Which of the following is correct with respect to the above statements?

- (A) Both statement (a) and statement (b) are true.
 (B) Both statement (a) and statement (c) are true.
 (C) Both statement (b) and statement (c) are true.
 (D) All the statements (a), (b) and (c) are true.

Q75 Consider a vocabulary with only four propositions A, B, C and D. How many models are there for the following sentence?

$$B \vee C$$

- (A) 10 (B) 12
 (C) 15 (D) 16

Q76 Consider the following statements :

- (a) $\text{False} \models \text{True}$
 (b) If $\alpha \models (\beta \wedge \gamma)$ then $\alpha = \beta$ and $\alpha \models \gamma$.

Which of the following is correct with respect to the above statements?

- (A) Both statement (a) and statement (b) are false.
 (B) Statement (a) is true but statement (b) is false.

- (C) Statement (a) is false but statement (b) is true.
 (D) Both statement (a) and statement (b) are true.

Q77 Consider the following English sentence :

“Agra and Gwalior are both in India”.

A student has written a logical sentence for the above English sentence in First-Order Logic using predicate $\text{In}(x, y)$, which means x is in y, as follows :

$$\text{In}(\text{Agra, India}) \vee \text{In}(\text{Gwalior, India})$$

Which one of the following is correct with respect to the above logical sentence ?

- (A) It is syntactically valid but does not express the meaning of the English sentence.
 (B) It is syntactically valid and expresses the meaning of the English sentence also.
 (C) It is syntactically invalid but expresses the meaning of the English sentence.
 (D) It is syntactically invalid and does not express the meaning of the English sentence.

Q78 Consider the following two sentences :

- (a) The planning graph data structure can be used to give a better heuristic for a planning problem.
 (b) Dropping negative effects from every action schema in a planning problem results in a relaxed problem.

Which of the following is correct with respect to the above sentences ?

- (A) Both sentence (a) and sentence (b) are false.
 (B) Both sentence (a) and sentence (b) are true.
 (C) Sentence (a) is true but sentence (b) is false.
 (D) Sentence (a) is false but sentence (b) is true.

Q79 A knowledge base contains just one sentence, $\exists x \text{ AsHighAs}(x, \text{Everest})$. Consider the following two sentences obtained after applying existential instantiation.

- (a) $\text{AsHighAs}(\text{Everest, Everest})$
 (b) $\text{AsHighAs}(\text{Kilimanjaro, Everest})$

Which of the following is correct with respect to the above sentences ?

- (A) Both sentence (a) and sentence (b) are sound conclusions.



- (B) Both sentence (a) and sentence (b) are unsound conclusions.
 (C) Sentence (a) is sound but sentence (b) is unsound.
 (D) Sentence (a) is unsound but sentence (b) is sound.

Q80 Consider the set of all possible five-card poker hands dealt fairly from a standard deck of fifty-two cards. How many atomic events are there in the joint probability distribution ?

- (A) 2, 598, 960 (B) 3, 468, 960
 (C) 3, 958, 590 (D) 2, 645, 590

Q81 E is the number of edges in the graph and f is maximum flow in the graph. When the capacities are integers, the runtime of Ford-Fulberson algorithm is bounded by :

- (A) $O(E * f)$ (B) $O(E^2 * f)$
 (C) $O(E * f^2)$ (D) $O(E^2 * f^2)$

Q82 Which of the following statements is false about convex minimization problem ?

- (A) If a local minimum exists, then it is a global minimum
 (B) The set of all global minima is convex set
 (C) The set of all global minima is concave set
 (D) For each strictly convex function, if the function has a minimum, then the minimum is unique

Q83 The following LPP

$$\text{Maximize } z = 100x_1 + 2x_2 + 5x_3$$

Subject to

$$14x_1 + x_2 - 6x_3 + 3x_4 = 7$$

$$32x_1 + x_2 - 12x_3 \leq 10$$

$$3x_1 - x_2 - x_3 \leq 0$$

$$x_1, x_2, x_3, x_4 \geq 0$$

has

- (A) Solution : $x_1 = 100, x_2 = 0, x_3 = 0$
 (B) Unbounded solution
 (C) No solution
 (D) Solution : $x_1 = 50, x_2 = 70, x_3 = 60$

Q84 Digital data received from a sensor can fill up 0 to 32 buffers. Let the sample space be

$S = \{0, 1, 2, \dots, 32\}$ where the sample j denote that j of the buffers are full and $p(i) = \frac{1}{561}(33 - i)$. Let A denote the event that the even number of buffers are full. Then p(A) is :

- (A) 0.515 (B) 0.785
 (C) 0.758 (D) 0.485

Q85 The equivalence of $\neg \exists x Q(x)$ is :

- (A) $\exists x \neg Q(x)$
 (B) $\forall x \neg Q(x)$
 (C) $\neg \exists x \neg Q(x)$
 (D) $\forall x \neg Q(x) \forall x Q(x)$

Q86 If $A_i = \{-i, \dots, -2, -1, 0, 1, 2, \dots, i\}$ then $\bigcup_{i=1}^{\infty} A_i$ is :

- (A) Z (B) Q
 (C) R (D) C

Q87 Match the following in List - I and List - II, for a function f :

List - I	List - II
(a) $\forall x \forall y (f(x) = f(y) \rightarrow x = y)$	(i) Constant
(b) $\forall y \exists x (f(x) = y)$	(ii) Injective
(c) $\forall x f(x) = k$	(iii) Surjective

- (A) a - (i),
 b - (ii),
 c - (iii)

- (B) a - (iii), b - (ii), c - (i)
 (C) a - (ii), b - (i), c - (iii)
 (D) a - (ii), b - (iii), c - (i)

Q88 Which of the relations on $\{0, 1, 2, 3\}$ is an equivalence relation ?

- (A) $\{(0, 0) (0, 2) (2, 0) (2, 2) (2, 3) (3, 2) (3, 3)\}$
 (B) $\{(0, 0) (1, 1) (2, 2) (3, 3)\}$
 (C) $\{(0, 0) (0, 1) (0, 2) (1, 0) (1, 1) (1, 2) (2, 0)\}$
 (D) $\{(0, 0) (0, 2) (2, 3) (1, 1) (2, 2)\}$

Q89 Which of the following is an equivalence relation on the set of all functions from Z to Z?

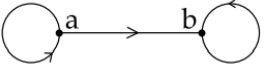
- (A) $\{(f, g) \mid f(x) - g(x) = 1 \forall x \in Z\}$
 (B) $\{(f, g) \mid f(0) = g(0) \text{ or } f(1) = g(1)\}$



- (C) $\{(f, g) \mid f(0) = g(1) \text{ and } f(1) = g(0)\}$
 (D) $\{(f, g) \mid f(x) - g(x) = k \text{ for some } k \in \mathbb{Z}\}$

Q90 Which of the following statements is true?

- (A) $(\mathbb{Z}, \leq)$ is not totally ordered
 (B) The set inclusion relation $\subseteq$ is a partial ordering on the power set of a set S
 (C) $(\mathbb{Z}, \neq)$ is a poset
 (D)

The directed graph  is not a partial order

Q91 CMOS is a Computer Chip on the motherboard, which is :

- (A) RAM
 (B) ROM
 (C) EPROM
 (D) Auxillary storage

Q92 In RS flip-flop, the output of the flip-flop at the time $(t+1)$ is the same as the output at time t , after the occurrence of a clock pulse if :

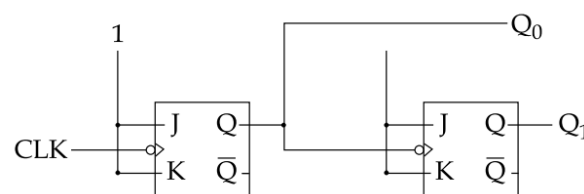
- (A) $S=R=1$ (B) $S=0, R=1$
 (C) $S=1, R=0$ (D) $S=R=0$

Q93 Match the terms in List - I with the options given in List - II :

List I	List II
(a) Decoder	(i) 1 line to 2^n lines
(b) Multiplexer	(ii) n lines to 2^n lines
(c) De-multiplexer	(iii) 2^n lines to 1 line
	(iv) 2^n lines to 2^{n-1} lines

- (A) (a)-ii (b)-i (c)-iii
 (B) (a)-ii (b)-iii (c)-i
 (C) (a)-ii (b)- i (c)- iv
 (D) (a)-iv (b)-ii (c)-i

Q94 What does the following logic diagram represent?



- (A) Synchronous Counter
 (B) Ripple Counter
 (C) Combinational Circuit
 (D) Mod 2 Counter

Q95 The hexadecimal equivalent of the binary integer number 110101101 is :

- (A) D24 (B) 1 B D
 (C) 1 A E (D) 1 A D

Q96 Perform the following operation for the binary equivalent of the decimal numbers

$$(-14)_{10} + (-15)_{10}$$

The solution in 8 bit representation is :

- (A) 11100011 (B) 00011101
 (C) 10011101 (D) 11110011

Q97 Match the items in List - I and List - II :

List- I	List - II
(a) Interrupts can be delayed when a much higher priority interrupt has occurred.	(i) Normal
(b) Unplanned interrupts which occur while executing a program.	(ii) Synchronous
(c) The source of the interrupt is in phase with the system clock	(iii) Maskable
	(iv) Exception

(A) (a) (b) (c)

(ii) (i) (iv)

(B) (a) (b) (c)

(ii) (iv) (iii)

(C) (a) (b) (c)

(iii) (i) (ii)

(D) (a) (b) (c)



(iii) (iv) (ii)

Q98 Which of the following mapping is not used for mapping process in cache memory ?

- (A) Associative mapping
- (B) Direct mapping
- (C) Set-Associative mapping
- (D) Segmented - page mapping

Q99 Simplify the following using K-map :

$$F(A,B,C,D) = \sum (0,1,2,8,9,12,13)$$

$$d(A,B,C,D) = \sum (10, 11,14,15)$$

d stands for don't care condition.

- (A) $A + B'D' + BC$
- (B) $A + B'D' + B'C'$
- (C) $A' + B'C'$
- (D) $A' + B'C' + B'D'$

Q100 In 8085 microprocessor, what is the output of following program?

LDA 8000 H

MVI B, 30 H

ADD B

STA 8001H

- (A) Read a number from input port and store it in memory
- (B) Read a number from input device with address 8000 H and store it in memory at location 8001 H
- (C) Read a number from memory at location 8000 H and store it in memory location 8001 H
- (D) Load A with data from input device with address 8000 H and display it on the output device with address 8001 H



Answer Key

Q1 (C)
Q2 (D)
Q3 (D)
Q4 (D)
Q5 (B)
Q6 (A)
Q7 (C)
Q8 (A)
Q9 (B)
Q10 (B)
Q11 (B)
Q12 (C)
Q13 (C)
Q14 (D)
Q15 (D)
Q16 (C)
Q17 (D)
Q18 (D)
Q19 (B)
Q20 (A)
Q21 (A)
Q22 (B)
Q23 (C)
Q24 (D)
Q25 (D)
Q26 (B)
Q27 (D)
Q28 (C)
Q29 (B)
Q30 (D)
Q31 (C)

Q32 (D)
Q33 (D)
Q34 (C)
Q35 (A)
Q36 (C)
Q37 (D)
Q38 (A)
Q39 (B)
Q40 (A)
Q41 (A)
Q42 (B)
Q43 (C)
Q44 (B)
Q45 (B)
Q46 (C)
Q47 (A)
Q48 (B)
Q49 (D)
Q50 (C)
Q51 (B)
Q52 (D)
Q53 (D)
Q54 (A)
Q55 (A)
Q56 (C)
Q57 (C)
Q58 (B)
Q59 (B)
Q60 (C)
Q61 (A)
Q62 (C)



Q63 (B)
Q64 (A)
Q65 (D)
Q66 (A)
Q67 (A)
Q68 (D)
Q69 (D)
Q70 (A)
Q71 (B)
Q72 (A)
Q73 (A)
Q74 (A)
Q75 (B)
Q76 (D)
Q77 (A)
Q78 (B)
Q79 (D)
Q80 (A)
Q81 (A)

Q82 (C)
Q83 (B)
Q84 (A)
Q85 (B)
Q86 (A)
Q87 (D)
Q88 (B)
Q89 (D)
Q90 (B)
Q91 (B)
Q92 (D)
Q93 (B)
Q94 (B)
Q95 (D)
Q96 (A)
Q97 (D)
Q98 (D)
Q99 (B)
Q100 (C)



Hints & Solutions

Q1 Text Solution:

Option- **C(valid)**

Explanation: A document is considered valid in XML if it follows the rules outlined in its related Document Type Definition (DTD) or XML Schema. This signifies that the document's structure and tags follow the definitions specified in the DTD or standard.

Q2 Text Solution:

Option- **D(M=2; D1 displays "12"; D2 displays "6")**

Explanation:

1. Global Variable: `var y = "12";` declares a global variable `y` with the value `"12"`.
2. Inside the Function `f`: `var y = "6";` declares a local variable `y` specific to the function `f`. This does not affect the global `y`.
3. First `alert` (D1): `alert(this.y);` refers to the global object (`this` in a browser points to `window`). `this.y` accesses the global `y`, so it displays `"12"`.
4. Second `alert` (D2): The function `g()` accesses the local `y` defined in `f`. `alert(y);` displays the value of the local `y`, which is `"6"`.

Q3 Text Solution:

Option- **D(15 : 15)**

Explanation:

1. **Array 'P':** Initially `{3, 4, 5}`.
2. **Method Call:** `method(P)` changes `P[1]` to `7`. Now `P = {3, 7, 5}`.
3. **Printing:**
 - `P[0] + P[1] + P[2] = 3 + 7 + 5 = 15` → Prints `15`.
 - `Q` is the same as `P`, so `Q[0] + Q[1] + Q[2] = 15` → Prints `15`.

Q4 Text Solution:

Option: **D(5, 2, 517)**

Explanation:

Step-by-Step:

1. `m.two[0] = 5`: The LSB is set to `5`.
2. `m.two[1] = 2`: The MSB is set to `2`.
3. `m.one` combines these bytes as per little-endian:

Endian Concept:

- Little-endian systems store the least significant byte (LSB) at the lowest and the most significant byte (MSB) at the higher memory address.
- For a 2-byte value like `short int`, the first byte (`m.two[0]`) is the LSB, and the second byte (`m.two[1]`) is the MSB.
 - Low byte: `5`, High byte: `2`.
 - Combined value: $2 \times 256 + 5 = 517$.
- `m.two[0] = 5`, `m.two[1] = 2`, `m.one = 517`.

Q5 Text Solution:

Option- **B((b) only)**

Explanation:

- (a) In this function, `a` and `b` are passed by value, which means the function works on copies of `p` and `q`. Any changes made to `a` and `b` inside the function will not affect the original values of `p` and `q` in `main()`. Therefore, this will not swap the contents of `p` and `q`.
 - (b) In this case, `a` and `b` are passed by reference (using `&`), meaning `a` and `b` directly refer to `p` and `q` from `main()`. The function swaps the values of `a` and `b`, which will reflect in `p` and `q`. This will correctly swap the contents of `p` and `q`.
 - (c) Here, `a` and `b` are pointers. The function swaps the values that `a` and `b` point to (using dereferencing `*a` and `*b`). The addresses of `p` and `q` are passed to the function, so the function modifies the original `p` and `q` directly.
- Conclusion:** (b) works correctly because it uses pass-by-reference, while (a) and (c) are either not modifying the values correctly or use pointer dereferencing in a way that does not meet the requirements for passing by reference.



Q6 Text Solution:Option- **A(S1 and S2 only)****Explanation:**

S1: True. When the `final` keyword is applied to a class, it prevents that class from being subclassed or extended. No class can extend a `final` class.

S2: True. Java supports single inheritance for classes (a class can only inherit from one superclass), but it supports multiple inheritance for interfaces (a class can implement more than one interface).

Q7 Text Solution:Option- **C(Both P and Q)****Explanation:**

P: C is frequently thought to have a weak type system since it allows for greater freedom in type conversions (such as casting between distinct types) without tight checks. Although C is statically typed (types are known at build time), the flexibility of type conversions and pointer arithmetic makes its type system less robust than languages that rigidly enforce type safety.

Q: Java is recognized for its strong type system, which requires stringent type checking. In Java, variables must be defined with precise types, and the compiler verifies type compliance. Java is also statically typed, which means that type checks are performed at compile time and before the program is executed.

Thus, Q is also correct.

Q8 Text Solution:Option- **A(16 frames per second)****Explanation:**

- Frame size = 640 pixels (width) x 480 pixels (height) = 307,200 pixels
- Access time per pixel = 200 nanoseconds
- Color depth = 1 bit (does not influence computation).

Step by Step:

1. Refresh time per frame: - Time for one pixel equals 200 nanoseconds.

- Total time to refresh all pixels = (307,200 x

200) nanoseconds = 61,440,000 nanoseconds = 61.44 milliseconds.

2. Calculate the refresh rate:

$$\text{Refresh rate} = \frac{1}{\text{Time for one frame in seconds}} = \frac{1}{61.44 \times 10^{-3}} \approx 16.3 \text{ frames per second}$$

Q9 Text Solution:Option- **B(S2 only)**

Explanation: Backface culling is a 3D rendering technique in which polygons facing away from the camera (known as "backfacing normals") are not rendered. This improves performance by ignoring geometry that isn't visible from the camera's perspective. A "backfacing" normal is a normal vector that points away from the camera.

Q10 Text Solution:

Option- **B(M represents first, a translation of vector (1, 1) followed by scaling of vector (2, 1))**

Explanation:

$$\text{Matrix } M: \begin{bmatrix} 2 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

The last column [2,1] represents a translation by vector (2,1).

The top-left submatrix $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ represents scaling:

Translation: The last column, represents the translation vector (tx,ty). Translation moves a point (x,y) by adding the translation vector:
Translated point:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

For M, tx=1, ty=1

Scaling: The top-left submatrix, $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$, represents scaling. Scaling modifies the size of an object:



Scaled point:

$$\begin{bmatrix} x'' \\ y'' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

For M, $s_x=2, s_y=1$:

$$\begin{bmatrix} x'' \\ y'' \\ 1 \end{bmatrix} = \begin{bmatrix} 2x \\ y \\ 1 \end{bmatrix}$$

The x-coordinate is scaled by 2.

The y-coordinate remains unchanged.

Combined Transformation

Applying M to translation :

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Q11 Text Solution:

Option- **B((54, 9, 27))**

Explanation: The correct calculations are:

Estimated Development Effort (E):

$$E = \frac{33,480 \text{ LOC}}{620 \text{ LOC/person-month}}$$

$$= 54 \text{ person-months}$$

Estimated Development Time (D):

$$D = \frac{54 \text{ person-months}}{6 \text{ developers}} = 9 \text{ months}$$

Estimated Development Cost (C):

$$C = \frac{6 \times 50,000 \times 9}{100,000} = 27 \text{ Lakhs}$$

Thus, (E, D, C) = (54, 9, 27).

Q12 Text Solution:

Option- **C((a) (b) (c) (d)**

(iv) (i) (ii) (iii)

Explanation:

	List I		List II	Explanation
(a)	Product Complexity	(iv)	Software Cost Estimation	Complexity impacts effort, time, and cost estimation for a product
(b)	Structured System	(i)	Software Requirements	Focuses on defining and

	Analysis		t Definition	analyzing software requirements systematically.
(c)	Coupling and Cohesion	(ii)	Software Design	These are key principles in designing modular and maintainable software.
(d)	Symbolic Execution	(iii)	Validation Technique	Used to validate and test code paths by executing with symbolic inputs.

Q13 Text Solution:

Option- **C(Syntax highlighting)**

Explanation: Text editors or Integrated

Development Environments (IDEs) are more likely to provide syntax highlighting than SCM software. SCM focuses on managing the codebase rather than altering it.

Q14 Text Solution:

Option- **D(99.9924%)**

Explanation: To calculate **software availability**, we use the formula:

$$\text{Availability} = \left(\frac{\text{Total Time} - \text{Downtime}}{\text{Total Time}} \right) \times 100$$

Step-by-Step Calculation:

1. Total Time in 2017: 1 year = $365 \times 24 \times 60$ minutes = 525,600 minutes.

2. Downtime:

Number of crashes = 20.

Each crash took 2 minutes to restart.

Total downtime = $20 \times 2 = 40$ minutes.

3. Available Time: Total time - Downtime =



$$525,600 - 40 = 525,560 \text{ minutes.}$$

4. Availability:

$$\text{Availability} = \left(\frac{525,560}{525,600} \right) \times 100$$

$$= 99.9924\%$$

Q15 Text Solution:

Option- **D((a) (b) (c) (d) (e)**
(iv) (v) (ii) (iii) (i))

Explanation: The correct sequence is:

	List I	List II	Explanation
(a)	Initial	(iv) There may not exist a plan or it may be abandoned.	Processes are unpredictable, poorly controlled, and reactive.
(b)	Repeatable	(v) There's a plan and people stick to it.	Processes are repeatable, and project management is established.
(c)	Defined	(ii) The plan for a project comes from a template for plans.	Processes are documented, standardized, and integrated into the organization.
(d)	Managed	(iii) The plan uses processes that can be measured quantitatively.	Processes are measured and controlled
(e)	Optimizing	(i) Processes are improved quantitatively and continually.	Processes focus on continuous improvement

Q16 Text Solution:

Option- **C(Q and R only)**

Explanation:

Q: Data coupling refers to the scenario in which

modules communicate by transferring data through parameters. When only the necessary data is transmitted (rather than the entire structure), the coupling is deemed low, which is ideal.

R: Stamp coupling arises when modules share a bigger data structure (for example, a record or object) but only use a subset of it. This sort of coupling is more limited than common coupling yet less desirable than data coupling.

Q17 Text Solution:

Option- **D(proxy)**

Explanation: A proxy design pattern is commonly used to limit or regulate access to an item. It serves as a placeholder or substitute for another object, controlling access to it. The proxy serves as a middleman, executing specific tasks (such as access control, logging, or lazy initialization) before forwarding the request to the actual object.

Q18 Text Solution:

Option- **D(P, Q and R only)**

Explanation:

P: Software re-engineering allows legacy systems to adapt to new requirements by changing their code or structure, making them more relevant to current demands.

Q: Re-engineering is frequently used to update software by updating to newer technologies, platforms, or paradigms, which improves scalability, compatibility, and performance.

R: One of the primary reasons for re-engineering is to rework and restructure software to increase maintainability, making it easier to understand, alter, and expand.

Q19 Text Solution:

Option- **B(Dynamic verification)**

Explanation: Dynamic Verification refers to testing and debugging through execution (e.g.,



running the program to find errors), which is prohibited in the cleanroom method. Instead, it focuses on static verification via rigorous reviews and proofs.

Q20 Text Solution:

Option- **A(P only)**

Explanation: Refactoring includes reorganizing old code without altering its usefulness. The goal is to make the code clearer, more efficient, and easier to maintain while guaranteeing that the product performs consistently from the user's perspective.

Examples include renaming variables, splitting a huge function into smaller ones, and simplifying code logic without adding or eliminating functionality.

Q21 Text Solution:

Option A is Correct

We can solve the recurrence relation

$T(m) = T\left(\frac{3m}{4}\right) + 1$ using the Master Theorem or recursion tree method.

Identifying the components of the recurrence

The recurrence has the form:

$$T(m) = aT\left(\frac{m}{b}\right) + f(m)$$

Where:

- $a=1$ (number of subproblems),
- $b=\frac{4}{3}$ (factor by which the problem size is reduced),
- $f(m)=1$ (the work done outside the recursive call).

Calculate the critical value $p=\log_b a$

$$p = \log_b(a) = \log_{\frac{4}{3}}(1) = 0$$

(using $\log_x(1) = 0$ for any base x).

Compare $f(m)$ to m^p

Here, $f(m)=1$ and $m^p=m^0=1$. Thus, $f(m)$ is asymptotically the same as m^p .

Apply the Master Theorem

When $f(m)$ is asymptotically equal to m^p , the solution to the recurrence is:

$$T(m) = \Theta(m^p \log m) = \Theta(\log m)$$

The solution is $\Theta(\log m)$.

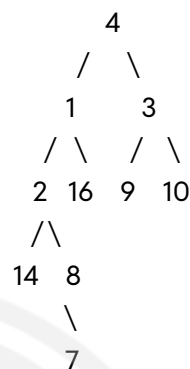
Q22 Text Solution:

Option B is Correct

Given the array $A = \langle 4, 1, 3, 2, 16, 9, 10, 14, 8, 7 \rangle$

1. Initial array representation:

- Index the array:
 $A[1]=4, A[2]=1, A[3]=3, \dots, A[10]=7$.
- This represents the following binary tree structure:



2. Max-Heapify the array: After applying the heap-building algorithm, we get the following max-heap:



Depth of the heap:

- The heap has $n=10$ elements.
- The depth (or height) of the heap is $\lfloor \log_2(10) \rfloor = 3$. (floor value)

Right child of the root:

- The root is 16.
- Its left child is 14, and its right child is 10.

Q23 Text Solution:

Option C is Correct

We are given:



- Hash Function: $h(\text{key}) = \text{key} \bmod 7$
- Collision Resolution: Linear Probing
- Keys to Insert: 44, 45, 79, 55, 91, 18, 63
- Table Index Range: 0 to 6

We need to determine the location of key 18 after inserting all the keys.

1. Insert key 44

$$h(44) = 44 \bmod 7 = 2$$

Key 44 is inserted at index 2.

2. Insert key 45

$$h(45) = 45 \bmod 7 = 3$$

Key 45 is inserted at index 3.

3. Insert key 79

$$h(79) = 79 \bmod 7 = 2$$

Index 2 is already occupied (by key 44). Using linear probing, the next available index is 4. Key 79 is inserted at index 4.

4. Insert key 55

$$h(55) = 55 \bmod 7 = 6$$

Key 55 is inserted at index 6.

5. Insert key 91

$$h(91) = 91 \bmod 7 = 0$$

Key 91 is inserted at index 0.

6. Insert key 18

$$h(18) = 18 \bmod 7 = 4$$

Index 4 is already occupied (by key 79). Using linear probing, check the next available indices and since, Index 5 is free.

Key 18 is inserted at index 5.

7. Insert key 63

$$h(63) = 63 \bmod 7 = 0$$

Index 0 is occupied (by key 91). Using linear probing, check the next available indices, and since Index 1 is free.

Key 63 is inserted at index 1.

Final Table After All Insertions:

Index	Key
0	91

Index	Key
1	63
2	44
3	45
4	79
5	18
6	55

Q24 Text Solution:

Option D is Correct

Dijkstra's Algorithm

Dijkstra's algorithm efficiently finds the shortest path from a single source to all other vertices in a graph with non-negative edge weights. It is explicitly designed to solve the single-source shortest paths problem.

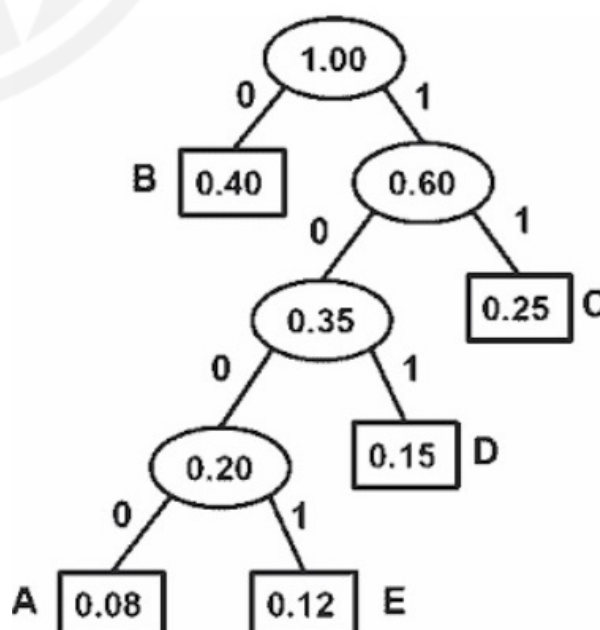
Q25 Text Solution:

Option D is Correct

To compute the average length of a Huffman code for the given probabilities of characters:

- A: 0.08
- B: 0.40
- C: 0.25
- D: 0.15
- E: 0.12

Build the Huffman Tree



Tree using Optimal Huffman Coding Technique



Sort probabilities in ascending order:

0.08, 0.12, 0.15, 0.25, 0.40

Combine the two smallest probabilities:

$$0.08 + 0.12 = 0.20$$

New list: 0.15, 0.20, 0.25, 0.40.

Combine the next two smallest probabilities:

$$0.15 + 0.20 = 0.35$$

New list: 0.25, 0.35, 0.40.

Combine the next two smallest probabilities:

$$0.25 + 0.35 = 0.60$$

New list: 0.40, 0.60.

Combine the last two probabilities:

$$0.40 + 0.60 = 1.00$$

The Huffman tree construction is complete.

Assign Binary Codes

Using the structure of the Huffman tree, assign codes to each character. The following tree gives us these codes:

- A: length = 4)
- B: length = 1)
- C: length = 2)
- D: length = 3)
- E: length = 4)

Compute the Average Length

The average code length is calculated as:

$$L_{avg} = \sum_i p_i l_i$$

Where p_i is the probability of the i -th character, and l_i is the length of its code.

$$L_{avg} = (0.08 \times 4) + (0.40 \times 1) + (0.25 \times 2) + (0.15 \times 3) + (0.12 \times 4)$$

Performing the calculations:

$$L_{avg} = (0.32) + (0.40) + (0.50) + (0.45) + (0.48) = 2.15$$

The computed average length is 2.15.

Q26 Text Solution:

Option B is Correct

A strictly binary tree (or full binary tree) is a binary tree in which every non-leaf node has exactly two children. For a strictly binary tree: $n = 2L - 1$ Where:

- n is the total number of nodes.
- L is the number of leaf nodes.

Given Data

- Number of leaves (L) = 19.
- We need to determine the total number of nodes (n).

Apply the Formula

Using the formula $n = 2L - 1$:

$$N = 2 \times 19 - 1 = 37$$

So, the tree has exactly 37 nodes.

Q27 Text Solution:

Option D is Correct

List-I	List-II	Explanation
(a) The 8-Queen's problem	(iv) Backtracking	The 8-Queens problem involves trying all possible placements and using backtracking to revert and try other possibilities when a solution path fails.
(b) Single-source shortest paths	(iii) Greedy approach	Algorithms like Dijkstra's use a greedy strategy to iteratively choose the shortest edge or path, ensuring optimality at every step.
(c) Strassen's Matrix Multiplication	(ii) Divide and conquer	Strassen's algorithm splits the matrix into smaller sub-matrices, solves them recursively, and combines the results, which is the divide-and-conquer paradigm.
(d) Optimal binary search trees	(i) Dynamic programming	Optimal binary search trees minimize search costs by solving subproblems (like cost of accessing subtrees) and combining results, which is characteristic of dynamic programming.

Q28 Text Solution:

Option C is Correct

Radix sort works by sorting each digit, one digit at a time, starting from the least significant digit (LSD) to the most significant digit (MSD). For each digit:

1. All n items are grouped into buckets (based on the current digit's value).



2. Sorting involves comparing and placing items for each digit.

Given:

1. Number of items (n): 9
2. Each item is a 5-digit octal number:
 - Octal numbers have base 8, meaning 8 possible values for each digit (0 to 7).
 - There are 5 digits in each number.

Number of Comparisons for Each Digit- For each digit, radix sort needs: $n \times b$ operations, where b is the base (number of possible values for each digit).

Here:

- $n=9$ (number of items),
- $b=8$ (octal base).

So, for one digit:

Comparisons for one digit = $n \times b = 9 \times 8 = 72$

Total Comparisons for 5 Digits, Since there are 5 digits, the total comparisons:

Total Comparisons = $5 \times$

(Comparisons for one digit)

Total Comparisons = $5 \times 72 = 360$

Q29 Text Solution:

Option B is Correct

In a complete n -ary tree:

$$L = (n-1) \cdot I + 1$$

Where:

- n : Number of children per internal node,
- I : Number of internal nodes,
- L : Number of leaf nodes.

Given:

- $n=5$ (5-ary tree),
- $I=8$ (number of internal nodes).

Since, $L = (n-1) \cdot I + 1$

Substitute the values:

$$L = (5-1) \cdot 8 + 1 = 33$$

NOTE: UGC NET has dropped the question.

Q30 Text Solution:

Option D is Correct

A Boolean function of n variables can be represented as:

$$f(x_1, x_2, \dots, x_n)$$

where $x_1, x_2, \dots, x_n \in \{0,1\}$.

Worst-Case Evaluation:

- To determine whether the function produces an output of 1, you may need to evaluate the function for all possible combinations of the n variables.
- The total number of combinations of n Boolean variables is 2^n .

This implies that the algorithm's complexity grows exponentially with n in the worst case.

Q31 Text Solution:

Option C is Correct

Finite State Machines (FSMs) are used to recognize patterns or tokens, and two FSMs are considered equivalent if they recognize the same language, i.e., they accept exactly the same set of input strings (tokens).

The structure of the FSMs (number of states or edges) does not matter as long as the set of strings (or tokens) they accept is the same. Even if two FSMs differ in their design (e.g., one has fewer states due to minimization), they can still be equivalent if they recognize the same language.

Q32 Text Solution:

Option D is Correct

The diagram shows:

- States that track the parity (odd/even) of a 's and b 's.
- The transitions between states depend on whether an a or b is encountered. The states represent:

1. Initial State: Even number of a 's and even number of b 's.
2. Transitions:
 - a : Changes the parity of a 's.
 - b : Changes the parity of b 's.



3. Accepting State: Odd number of a's and odd number of b's.

The FSM recognizes strings with odd number of a's and odd number of b's.

Q33 Text Solution:

Option D is Correct

A Pushdown Automaton (PDA) is a type of computational model that uses a stack as its auxiliary memory. However, a Turing Machine (TM) has unrestricted memory and is therefore more powerful than a PDA. For a PDA to simulate a Turing machine, it must have enough memory to handle arbitrary computation.

1. PDA with One Stack: A PDA with 1 stack can recognize only context-free languages (CFLs). It is strictly less powerful than a Turing machine because a single stack cannot emulate the unrestricted memory of a Turing machine.
2. PDA with Two Stacks: A PDA with 2 stacks can simulate a Turing machine:

- One stack can be used to simulate the tape of the Turing machine moving left.
- The other stack can simulate the tape moving right.

With 2 stacks, the PDA can recognize recursively enumerable languages (REs), the same as a Turing machine.

3. PDA with More than 2 Stacks:

Adding more than 2 stacks does not provide additional computational power beyond a Turing machine.

Q34 Text Solution:

Option C is Correct

A Pushdown Automaton (PDA) is a computational model that recognizes context-free languages (CFLs). However, since regular languages (recognized by finite automata) are a subset of context-free languages, a PDA can also recognize regular languages.

Q35 Text Solution:

Option A is Correct

In CNF, every production rule is in one of the following forms:

- $A \rightarrow BC$, where A, B, and C are non-terminal symbols.
- $A \rightarrow a$, where A is a non-terminal and a is a terminal.

Each production of the form $A \rightarrow BC$ generates two new non-terminals, and each terminal a corresponds to one final production.

1. To derive a string with n terminals, the CNF grammar must produce exactly n terminals in the final step.
2. Non-terminal productions: For n terminals, you need n-1 intermediate non-terminal productions to create the full binary structure that leads to n leaves (terminals).

This is because every $A \rightarrow BC$ production reduces the number of non-terminals by 1, and we start with n-1 such reductions.

3. Terminal productions: Each of the n terminals requires one terminal production ($A \rightarrow a$).

Total Productions = (n-1) non-terminal productions + n terminal productions = $2n-1$.

Q36 Text Solution:

Option C is Correct

A grammar is ambiguous if a string generated by it can have two or more distinct parse trees (or leftmost derivations).

Analyzing G1:

The grammar:

$S \rightarrow SbS \mid a$

Example String: ababa

1. ababa can be derived as follows:

- First derivation:
 $S \rightarrow SbS \rightarrow abS \rightarrow abSbS \rightarrow ababS \rightarrow ababa$
- Second derivation:
 $S \rightarrow SbS \rightarrow SbSbS \rightarrow abSbS \rightarrow ababS \rightarrow ababa$

2. Clearly, the string ababa can have multiple distinct parse trees.

Thus, G1 is ambiguous.



Analyzing G2:

The grammar:

$S \rightarrow aB \mid ab, A \rightarrow GAB \mid a, B \rightarrow ABb \mid b$

Example String: ab

1. ab can be derived as follows:

- First derivation: $S \rightarrow ab$
- Second derivation: $S \rightarrow aB, B \rightarrow b, S \rightarrow aB \rightarrow ab$

2. Since the string ab can be derived using multiple distinct parse trees, G2 is also ambiguous.

Hence, G1: Ambiguous (as shown with ababa), and G2: Ambiguous (as shown with ab).

Q37 Text Solution:

Explanation:

Context-sensitive language can be recognized by a Linear bounded automata.

An LBA is a type of Turing machine that operates with a tape of limited size, which is proportional to the length of the input.

The Final Answer is: Option D

Q38 Text Solution:

Explanation:

Context-sensitive grammars (CSGs) can handle dependencies within a string.

They ensure that the count of 0's, 1's, and 2's in a string are equal.

This ability makes CSGs essential for generating such strings.

Example:

$A = \{ 0^n 1^n 2^n \mid n = 1, 2, 3, \dots \}$

if $n=1$, then 012

if $n=2$, then 001122

if $n=3$, then 000111222, etc.

The Final Answer is: Context-sensitive

Q39 Text Solution:

Explanation:

A bottom-up parser generates the Right-most derivation in reverse.

A bottom-up parser starts with the input string(leaves of the parse tree).

Bottom-up parsing is also known as shift-reduce parsing because it involves shifting input symbols and reducing them to non-terminals based on grammar rules.

The Final Answer is: Option B

Q40 Text Solution:

Explanation:

S_1 : It is impossible to decide whether two Turing machines accept the same language, which is called the language equivalence problem, and no universal algorithm can solve it for all Turing machines.

S_2 : The Halting problem states that there's no algorithm to determine whether a Turing machine will halt on every possible input.

The Final Answer is that both S_1 and S_2 are correct

Q41 Text Solution:

Explanation:

According to Question Given information:

Frame size= 200 bits

Bandwidth= 200 Kbps

Frame rate= 250 frames per second

Step 1: Transmission time of one frame=

$$\frac{\text{Frame size}}{\text{Bandwidth}} = \frac{200 \text{ bits}}{200,000 \text{ bits/second}} = 1 \text{ millisecond (ms)}$$

Step 2: In slotted ALOHA, the time is divided into slots, and each station sends its frame at the beginning of a time slot. The system is unsuccessful if more than one station transmits during the same slot, resulting in a collision.

The Probability of a successful transmission(S) = $G \cdot e^{-G}$, where G is the average number of frame attempts per time slot.

Step 3: since the system produces 250 frames per second, and each frame takes 1 ms to transmit, the frame arrival rate G is:

$$G = \text{Frame rate} \times \text{Transmission time per frame} = 250 \text{ frames/sec} \times 1 \text{ ms} = 250 \times 0.001 = 0.25$$

$$\text{Step 4: The Probability of a successful transmission}(S) = G \cdot e^{-G} = 0.25 \times e^{-0.25} \approx 0.25 \times 0.778 \approx 0.1947$$



Now, Throughput = Frame rate $\times$ The Probability of a successful transmission(S)
 $= 250 \times 0.1947 \approx 48.6$ frames/sec
 ≈ 49 frames/sec

The Final Answer is: 49

Q42 Text Solution:

Explanation:

$f = \frac{1}{T}$ (where f is the frequency in Hertz(Hz), and T is the period in seconds(s))

Period(T) = 100 ms = 100×10^{-3} s = 0.1 sec

Now, $f = \frac{1}{0.1} = 10\text{Hz} = 10^{-2}$ KHZ (where 1 KHZ = 1000 Hz).

The Final Answer is: 10^{-2} KHZ

Q43 Text Solution:

Explanation:

10000001	00001011	00001011	11101111
First octet	Second octet	Third octet	Fourth octet

Convert each octet from Binary to decimal :

$10000001 = 1 \times 2^7 + 0 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 129$ (Decimal)

$00001011 = 0 \times 2^7 + 0 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 11$ (Decimal)

$00001011 = 0 \times 2^7 + 0 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 11$ (Decimal)

$11101111 = 1 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 239$ (Decimal)

The dotted-decimal notation is: 129.11.11.239

The Final Answer is: 129.11.11.239

Q44 Text Solution:

Explanation:

IS-95 is a second-generation cellular phone system based on CDMA and DSSS.

[IS-95 (CDMAOne) is a 2G system that uses CDMA and DS]

The third-generation cellular phone system will provide universal personal communication.

[3G systems provide global coverage and support multimedia services like video calls and mobile internet.]

AMPS is a first-generation system, not a second-generation system.

The Final Answer is: (b) and (c) only

Q45 Text Solution:

Explanation:

	List - I		List - II
(a)	DES	(ii)	block size 64 and key size 64.
(b)	IDEA	(iv)	block size 64 and key size 128.
(c)	BLOWFISH	(i)	block size 64 and key size ranges between 32 and 448.
(d)	AES	(iii)	block size 128 and key sizes 128, 192, 256.

The Final Answer is: a- ii, b-iv, c-i, d-iii

Q46 Text Solution:

Explanation:

True Statement:

(a) Three broad categories of Networks are

(i) Circuit Switched Networks (Example:

Traditional telephone systems)

(ii) Packet Switched Networks (Example: The Internet)

(iii) Message Switched Networks

(c) In packet-switched networks, resources are not reserved for each packet. Each packet is sent independently and may take different paths through the network.

The Final Answer is: (a) and (c) only

Q47 Text Solution:

Explanation:

In Challenge-Response Authentication, the person trying to prove their identity shows they know a secret (like a password) without actually giving it away. The system asks a question (the challenge), and the person answers using the secret to confirm who they are.

The Final Answer is: Option A

Q48 Text Solution:

Explanation:

Ciphertext: W T A A D

Shift each letter back by 15 positions:



Ciphertext	Letter Position	Shifting back	Decrypt Message	Position
W	23	15	H	8
T	20	15	E	5
A	1	15	L	12
A	1	15	L	12
D	4	15	O	15

Decrypt Message: HELLO

The Final Answer is Option B.

Q49 Text Solution:

Explanation:

The formula of the minimum Hamming distance needed to correct t errors is $\rightarrow d_{\min} = 2t + 1$

The Final Answer is: Option D

Q50 Text Solution:

Explanation:

Given Message = HELLO MY DEARZ

Block size = 4

Original $\rightarrow$ Cipher text: 1 2 3 4

1	2	3	4
H	E	L	L
O	M	Y	D
E	A	R	Z

After applying on key (Plain text: 2 4 1 3)

2	4	1	3
E	L	H	L
M	D	O	Y
A	Z	E	R

The Final Answer is: ELHL MDOY AZER

Q51 Text Solution:

Explanation:

Step 1: The initial value of the semaphore equals 10.

Step 2: After 12 P operations $\rightarrow$ (Each P operation decreases the semaphore value by 1)
After 12 P operations, the semaphore value is $10 - 12$ equals -2 .

Step 3: After 'x' V operations $\rightarrow$ (Each V operation increases the semaphore value by 1)

After 'x' V operations, the semaphore value is $-2 + x$.

Step 4: Final value of semaphore is 7 equals $-2 + x$

$$\Rightarrow 7 = -2 + x$$

$$\Rightarrow x = 9$$

The Final Answer is: 9

Q52 Text Solution:

Explanation:

Given Information:

Hit Ratio = 0.40

Access time for Primary memory = 15 ns

Access time for Secondary Memory = 120 ns

Miss Ratio = $1 - \text{Hit Ratio} = 1 - 0.40 = 0.60$

Effective average access time = (Hit ratio $\times$ Access time for Primary memory) + (Miss ratio $\times$ Access time for secondary memory)

$$\Rightarrow (0.40 \times 15) + (0.60 \times 120)$$

$$\Rightarrow 6 + 72$$

$$\Rightarrow 78$$

The average time required to access a page is 78.

The Final Answer is: Option D

Q53 Text Solution:

Poisson distribution is used for modeling the number of events occurring within a fixed interval of time when the events happen independently and at a constant average rate. Adjust the period to match the rate of requests.

40 minutes is $\frac{2}{3}$ of an hour, so in hours:

$$t = 40 \text{ minutes} = \frac{40}{60} = \frac{2}{3} \text{ hours.}$$

Adjust the average rate for 40 minutes

The average number of requests in 1 hour is **30 requests per hour**.

For **40 minutes** ($\frac{2}{3}$ of an hour), the expected number of requests is:

$$\lambda = 30 \times \frac{2}{3} = 20.$$

Expected number of requests in 40 minutes is **20**.

Poisson distribution

The Poisson distribution formula for the probability of observing **k events** (in this case, 0 events):



$$P(k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

where λ is the expected number of events, and k is the number of events.

Probability that **no requests are made in 40 minutes**, $k=0$:

$$P(0) = \frac{20^0 e^{-20}}{0!} = e^{-20}$$

The probability that **no requests** are made in 40 minutes is:

$$P(0) = e^{-20}$$

Thus, the correct answer is **D: e^{-20}** .

Q54 Text Solution:

In a **memory-mapped I/O system**, the operating system controls access to memory regions that correspond to I/O devices.

The operating system can ensure **protection** by **validating** and **restricting access** to certain memory areas to prevent user programs from accessing hardware resources directly.

For example, the OS can enforce access rights by controlling which processes can interact with certain memory-mapped I/O addresses.

Correct: I/O protection in memory-mapped I/O systems is managed by the operating system through access control mechanisms.

Q55 Text Solution:

Option A: `chmod -R a+x progs`

Correct:

`chmod`: This command is used to change file permissions.

`-R`: The recursive option, means the command will apply to all files and subdirectories within the specified directory.

`a+x`: To add the execute permission (x) for all users (a stands for all users — owner, group, and others).

`progs`: This is the target directory where the permissions are to be changed.

Result: This command will make all files and

subdirectories in "progs" executable by all users (owner, group, and others).

Q56 Text Solution:

Statement (b):

Memory Fragmentation can be internal as well as external.

- This statement is true.
- Memory fragmentation can indeed be both internal and external:
 - Internal Fragmentation occurs when allocated memory may have unused space because the allocated block is larger than the required size.
 - External Fragmentation occurs when free memory is available, but it is scattered across small blocks and not contiguous enough to satisfy a request.

Statement (c):

One solution to external fragmentation is compaction.

- Compaction is a technique used to combat external fragmentation. It involves moving processes around in memory to create larger contiguous blocks of free memory. This process reduces fragmentation by consolidating free space and making it possible to allocate larger memory blocks to processes that need them.

This statement is correct.

Q57 Text Solution:

The essential content in each entry is the Page Frame Number (PFN), which represents the physical memory location (frame) where a page from virtual memory is stored.

Page Table Structure:

The Page Table essentially maps virtual addresses to physical addresses.

Each entry in the page table corresponds to a virtual page and contains the Page Frame Number (PFN), which is the actual location in physical memory where the page resides.



This mapping helps the operating system locate where to fetch data from or write data in physical memory when a process accesses a virtual address.

The **Page Frame Number** is crucial because it directly maps a virtual page to a physical page in memory.

Q58 Text Solution:

Reference string: 1, 2, 3, 2, 4, 2, 5, 2, 3, 4

Number of frames = 3

	1	2	2	2	2
	2	3	4	5	3
	3	4	5	3	4
Page Fault	3	1	1	1	1

Here, total page faults = 3+1+1+1+1=7

Q59 Text Solution:

Option B is Correct

Given Processes:

- P₁: Arrival Time = 0 ms, Burst Time = 7 ms
- P₂: Arrival Time = 1 ms, Burst Time = 4 ms
- P₃: Arrival Time = 2 ms, Burst Time = 8 ms

In Preemptive SJF, the CPU is allocated to the process with the shortest remaining burst time at any given time. If a process with a shorter burst time arrives while another process is executing, it preempts the currently executing process.

Timeline Execution

Time	Executing Process	Remaining Burst Times (P ₁ , P ₂ , P ₃)	Explanation
0–1	P ₁	P ₁ =6, P ₂ =4, P ₃ =8	P ₁ starts execution as it arrives first.
1–5	P ₂	P ₁ =6, P ₂ =0, P ₃ =8	P ₂ arrives and preempts P ₁ (shorter burst time).
5–11	P ₁	P ₁ =0, P ₃ =8	P ₂ finishes, P ₁ resumes (shorter remaining time than P ₃).
11–19	P ₃	P ₃ =0	P ₁ finishes, P ₃ starts and completes.

Gantt Chart

The Gantt Chart for the execution is as follows:

P ₁	P ₂	P ₁	P ₃
0			
1	5	11	19

Q60 Text Solution:

Non-preemptive SJF is a non-preemptive scheduling algorithm.

- Once a process starts executing, it runs to completion without being interrupted.
- No context switching occurs unless a process completes or the CPU is idle and there are new processes to execute.
- A running process will not be preempted by another process, so there is no context switching between processes.

Q61 Text Solution:

Inner Join:

The **Inner Join** returns all **rows from both tables** where there is a match based on the join condition.

If there is **no match** between the tables based on the condition, the rows will **not** be included in the result.

Inner Join returns **only the rows** that satisfy the join condition

Q62 Text Solution:

SELECT COUNT(*)

FROM book AS T

WHERE T.price > B.price

A subquery is for each book B in the outer query. It counts how many books in the table (book) have a price greater than the price of the book B.

Condition:

WHERE (SELECT COUNT(*) FROM book AS T WHERE T.price > B.price) < 7;

Include book B if there are fewer than 7 books that have a price greater than the price of B.

For the book with the highest price:

There are zero books with a higher price, so COUNT(*) = 0.

Since 0 < 7, this book will be included.

For the book with the second highest price:



There is 1 book with a higher price, so $\text{COUNT}^* = 1$.

Since $1 < 7$, this book will be included.

Similarly, for the third most expensive book, $\text{COUNT}^* = 2$, and it will be included.

For the sixth most expensive book, $\text{COUNT}^* = 5$, it will be included.

For the seventh most expensive book, $\text{COUNT}^* = 6$, it will be included.

For the eighth most expensive book, $\text{COUNT}^* = 7$, it won't be included because $7 < 7$ is false.

C: Titles of the seven most expensive books.

Q63 Text Solution:

In hierarchical databases, the **root** of the hierarchy is the starting point, from which all other records (child records) are linked. In some cases, a **hashing function** is used to **locate the root** more efficiently, especially when the structure is large or distributed. The root is the most important entry in the hierarchical structure because it serves as the entry point to all other data.

Hashing for Root: A hashing function can be applied to a key associated with the **root** record to find its location in memory or storage more quickly. Once the root is located, traversing the hierarchical structure (children, grandchildren, etc.) be done more efficiently.

Q64 Text Solution:

When relations are produced from an **Entity-Relationship (E-R) model**, they are in **1NF** because:

1NF requires that all attributes contain only atomic values (no multi-valued or composite attributes). In the E-R model, entities are represented as sets of attributes, and relations derived from these entities must ensure that all attributes are atomic by definition.

Example:

A relation will not have sets, lists, or arrays as values in any attribute.

Multi-valued attributes are handled by creating new relations (i.e., breaking them down into

simpler relations), ensuring atomic values in the resulting tables.

When converting an E-R model to a relational model, the **relations** (tables) will always satisfy **1NF**, since the E-R model inherently ensures that attributes are atomic.

Option A is correct.

Q65 Text Solution:

Schedule S_1 :

$r_1(X); r_1(Y); r_2(X); r_2(Y); w_2(Y); w_1(X)$

$r_1(X)$ and $r_2(X)$: These are read operations on X.

No conflict here.

$r_1(Y)$ and $r_2(Y)$: These are read operations on Y.

No conflict here.

$w_2(Y)$ and $r_1(Y)$: This is a write-read conflict because transaction 2 writes to Y and transaction 1 reads Y. Conflict occurs.

$w_1(X)$ and $r_2(X)$: This is a write-read conflict because transaction 1 writes to X and transaction 2 reads X. Conflict occurs.

Schedule S_2 :

$r_1(X); r_2(X); r_2(Y); w_2(Y); r_1(Y); w_1(X)$

$r_1(X)$ and $r_2(X)$: These are read operations on X.

No conflict here.

$r_1(Y)$ and $r_2(Y)$: These are read operations on Y.

No conflict here.

$w_2(Y)$ and $r_1(Y)$: This is a write-read conflict because transaction 2 writes to Y and transaction 1 reads Y. Conflict occurs.

$w_1(X)$ and $r_2(X)$: This is a write-read conflict because transaction 1 writes to X and transaction 2 reads X. Conflict occurs.

Conflict Graph for S_2 :

In this case, the conflict graph does not contain a cycle, meaning S_2 is conflict serializable.



Option D is the correct answer.

Q66 Text Solution:

1NF (First Normal Form) requires that all attributes in the relation have atomic values. The



question states that the domains of **a**, **b**, **c**, and **d** include only atomic values. Therefore, we assume that the relation is in **1NF**.

Candidate keys

The candidate keys are the minimal set of attributes that uniquely identify a tuple in the relation.

a → c: **a** uniquely determines **c**.

b → d: **b** uniquely determines **d**.

For the relation to be uniquely identified, we need both **a** and **b** together because neither **a** nor **b** alone can identify the relation uniquely (since **a** determines only **c** and **b** determines only **d**).

So, the **candidate key** for this relation is **{a, b}**.

Check if the relation is in 2NF

2NF (Second Normal Form) requires that the relation is in **1NF** and there are no **partial dependencies**. A partial dependency occurs when a non-prime attribute (an attribute that is not part of the candidate key) is functionally dependent on a **proper subset** of any candidate key.

a → c is a partial dependency because **a** is part of the candidate key **{a, b}**. The partial dependency violates **2NF** because **c** depends on just part of the candidate key (i.e., on **a**, which is a subset of **{a, b}**).

Therefore, the relation **does not satisfy 2NF** because of the partial dependency **a → c**.

Option A is correct.

Q67 Text Solution:

In a **many-to-one** relationship between two entity sets **r₁** and **r₂**, each entity in **r₂** is associated with at most one entity in **r₁**, but an entity in **r₁** is associated with multiple entities in **r₂**.

Primary key of r₁ determines the **primary key of r₂**:

Relationship is **many-to-one**, each entity in **r₂** is associated with exactly one entity in **r₁**, but the reverse is not true. **Primary key of r₁ (Pkr₁)** can uniquely identify an entity in **r₂** (i.e., each entity in **r₂** is mapped to a single entity in **r₁**).

This relationship is captured by the functional dependency: **Pk(r₁) → Pk(r₂)**, meaning the primary key of **r₁** functionally determines the primary key of **r₂**.

Q68 Text Solution:

The OS is designed for fast and efficient file access control, which can speed up the process of granting and revoking permissions. As the authorization procedures are offloaded to the OS, the database system doesn't need to manage its complex security mechanisms, which can improve overall performance.

Option D is correct.

Q69 Text Solution:

Operation A (Insert into R1) and Operation D (Delete from R2) are the operations that can violate the referential integrity constraint:

- Insert into R1 may insert a foreign key value that doesn't exist in R2.
- Delete from R2 may delete a primary key value that R1 is referencing, thus breaking the foreign key constraint.

Q70 Text Solution:

For 1:

- Hash function: $(7 * 1 + 3) \bmod 4$
- Result: $(7 + 3) \bmod 4 = 10 \bmod 4 = 2$
- So, 1 will go into index 2 of the table.

For 3:

- Hash function: $(7 * 3 + 3) \bmod 4$
- Result: $(21 + 3) \bmod 4 = 24 \bmod 4 = 0$
- So, 3 will go into index 0 of the table.

For 8:

- Hash function: $(7 * 8 + 3) \bmod 4$
- Result: $(56 + 3) \bmod 4 = 59 \bmod 4 = 3$
- So, 8 will go into index 3 of the table.

For 10:

- Hash function: $(7 * 10 + 3) \bmod 4$
- Result: $(70 + 3) \bmod 4 = 73 \bmod 4 = 1$
- So, 10 will go into index 1 of the table.



After inserting the sequence **1, 3, 8, 10** into the hash table, the content of the table will be:

Index	Value
0	3
1	10
2	1
3	8
4	—
5	—
6	—

Q71 Text Solution:

In Artificial Intelligence (AI), an environment is considered uncertain if:

- It is not fully observable (i.e., the agent does not have access to all the information about the environment at any given time).
- It is not deterministic (i.e., the outcome of actions is not predictable and may vary).

If an environment is either not fully observable or not deterministic, it introduces uncertainty, as the agent may not have complete knowledge or cannot predict the exact consequences of its actions. This makes decision-making more challenging.

Q72 Text Solution:

In Artificial Intelligence (AI), a simple reflex agent selects actions based on the **current percept, completely ignoring the rest of the percept history**. This means it makes decisions solely based on the current situation or environment without considering past experiences.

Q73 Text Solution:

In heuristic search algorithms in Artificial Intelligence (AI), if we have a collection of admissible heuristics ($h_1, h_2, \dots, h_m$) for a problem and none of them dominates the others, we should choose:

$$h(n) = \max \{ h_1(n), \dots, h_m(n) \}$$

This approach ensures that the heuristic function ($h(n)$) is also admissible and provides the most

informed estimate by taking the maximum value among the available heuristics.

Q74 Text Solution:

To solve this question, let's analyze each statement about the A* algorithm:

- (a) A* expands all nodes with $f(n) < C^*$: This is true. A* will expand all nodes where the estimated cost ($f(n)$) is less than the cost of the optimal solution path (C^*).
- (b) A* expands no nodes with $f(n) \geq C^*$: This is also true. A* will not expand nodes where the estimated cost ($f(n)$) is greater than or equal to the cost of the optimal solution path (C^*).
- (c) Pruning is integral to A*: This statement is not necessarily true. While A can benefit from pruning to improve efficiency, pruning is not an inherent part of the A algorithm itself.

Based on this analysis, the correct answer is:

Both statement (a) and statement (b) are true.

Q75 Text Solution:

Sentence: $B \vee C$

Truth Table:

	A	B	C	D	B V C
	T	T	T	T	T
	T	T	T	F	T
	T	T	F	T	T
	T	T	F	F	T
	T	F	T	T	T
	T	F	T	F	T
	T	F	F	T	F
	T	F	F	F	F
	F	T	T	T	T
	F	T	T	F	T
	F	T	F	T	T
	F	T	F	F	T
	F	F	T	T	T
	F	F	T	F	T
	F	F	F	T	F



				B
A	B	C	D	∨
				C
F	F	F	F	F

12 models are there for given Sentence.

Tips and Tricks:

Number of variables = 4

Statement possible = $2^4 = 16$

Since $(B \vee C)$ OR operation between two

variables $\therefore \frac{3}{4}$ of total cases will be true.

Therefore, number of modes = $\frac{3}{4} \times 16 = 12$

Q76 Text Solution:

Explanation

Key Idea of $\models$ (Entailment):

- **Semantic relationship:** $A \models B$ means that if A is true, B must also be true in all situations.
- **If A is false:** If A is false, it can still entail any conclusion, including B , because a false premise cannot disprove the conclusion.

Cases:

1. Case 1: True $\models$ True

If the premise is true, the conclusion must also be true. This is a valid entailment.

2. Case 2: False $\models$ True

If the premise is false, it can logically entail any conclusion (True or False), because there's no situation where the premise can be true to disprove the conclusion.

3. Case 3: True $\models$ False

If the premise is true, it cannot lead to a false conclusion. This is not valid.

4. Case 4: $\alpha \models (\beta \wedge \gamma)$

If α logically entails both β and γ , then α must logically entail γ as well.

Solution :

Statement (a): False $\models$ True

- False $\models$ True holds true because when the premise is false, the entailment rule says it is automatically true.

- There is no contradiction because the premise cannot be true to test against the conclusion. This is called **vacuous truth** in logic.

Conclusion: Statement (a) is **True**.

Statement (b): If $\alpha \models (\beta \wedge \gamma)$, then both $\alpha \models \beta$ and $\alpha \models \gamma$.

• Given: $\alpha \models (\beta \wedge \gamma)$

- This means that whenever α is true, both β and γ must also be true.

• Proof of $\alpha \models \beta$:

- For $\alpha \models (\beta \wedge \gamma)$, α must logically entail both β and γ .
- It is reasonable to assume that α and β are closely connected in this case, which suggests that $\alpha \models \beta$ (i.e., α and β must have the same truth value).

• Proof of $\alpha \models \gamma$:

- Since $\alpha \models (\beta \wedge \gamma)$, and $\beta \wedge \gamma$ is true only if both β and γ are true, it follows that if α is true, γ must also be true.
- Therefore, $\alpha \models \gamma$ holds.

Conclusion: Statement (b) is **True**.

So, Option D is correct .

Q77 Text Solution:

The given statement is "Agra and Gwalior are both in India".

Predicate $In(x, y)$ means x is in y .

So,

1) $In(\text{Agra}, \text{India})$ — It represents Agra in India.

2) $In(\text{Gwalior}, \text{India})$ — It represents Gwalior in India.

Given predicate logic means

$In(\text{Agra}, \text{India}) \vee In(\text{Gwalior}, \text{India})$: It represents Agra is in India or Gwalior is in India.

It does not represent that both Agra and Gwalior are in India.

So, we can say that It is syntactically valid but



does not express the meaning of the English sentence.

Q78 Text Solution:

(a) Planning Graph for Heuristic

A planning graph provides a better heuristic by representing time steps as levels, starting from the initial state. If a literal appears in the final level, it is achievable; if not, it's not possible to achieve. The level at which a goal literal appears can estimate the cost to achieve it, improving heuristic accuracy.

(b) Dropping Negative Effects

Removing negative effects from action schemas results in a relaxed problem, making planning simpler. Negative effects impose restrictions that increase time complexity, so eliminating them increases possible actions, leading to a less constrained and faster solution process.

So, option (B) is correct.

Q79 Text Solution:

Concept:

Predicate logic deals with predicates which are propositions, consist of variables.
Two quantifiers in predicate logic are there exist ($\exists$) and for all ($\forall$).

Explanation:

Given statement is:

$\exists x \text{ AsHighAs } (x, \text{Everest}).$

It is comparing one element with another element. It means, there exists a element x which is as high as Everest. So, for comparison we have x and Everest should be different.

(a) AsHighAs (Everest, Everest)

After applying existential quantifies, this statement becomes.

$\exists \text{Everest AsHighAs } (\text{Everest}, \text{Everest})$: it means there exist an Everest which is as high as Everest. There is no meaning of this sentence. As we are

comparing same elements. It does not sounds good.

(b) AsHighAs (Kilimanjaro, Everest)

After applying existential quantifies, this statement becomes.

$\exists \text{Killimanjaro AsHighAs } (\text{Killimanjaro}, \text{Everest})$: it means there exists a Killimanjaro which is as high as Everest. It is comparing Killimanjaro with Everest which does not have any problem. It sounds good.

Q80 Text Solution:

Concept:

A joint probability distribution shows a probability distribution for two or more random variables. It looks for a relationship between two variables say x and y. It gives the probability that each x and y falls in a particular range of set of values specified for that variable.

Explanation:

Here, given:

Total cards = 52

We have to deal with 5 cards.

In this question, atomic events possible can be find in the number of ways they can be combined.

Which is represented as $C(n, r) = \frac{n!}{(n-r)! r!}$

Here $n = 52$

$r = 5$

$C(52, 5) = \frac{52!}{(52-5)! 5!} = \frac{52!}{47! 5!}$

$C(52, 5) = \frac{52 \times 51 \times 50 \times 49 \times 48 \times 47!}{5 \times 4 \times 3 \times 2 \times 1 \times 47!}$

$C(52, 5) = 2, 598, 960$

Q81 Text Solution:

When the capacities are integers, the runtime of the Ford-Fulkerson algorithm is bounded by $O(E \cdot f)$, where (E) is the number of edges in the graph and (f) is the maximum flow in the graph. This is because each augmenting path can be found in $O(E)$ time and increases the flow by at least one unit, with the maximum flow being (f).

NOTE: UGC NET has dropped this question.



Q82 Text Solution:

Some points about convex problems are:

- With a convex objective and convex feasible region, there can be only one optimal solution which is globally optimal.
- convex problem can be solved efficiently up to large size
- There are two methods to solve convex problems: interior point or barrier methods.
- Every local minimum exists in the problem is also a global minimum.
- Set of all global minima is a convex set.
- Set of all global maxima is a concave set
- For each strictly convex function, if the function has a minimum, then the minimum is unique.

Q83 Text Solution:

Objective:

Maximize $z = 100x_1 + 2x_2 + 5x_3$

Step 1: Convert the inequalities into equalities by introducing slack variables:

We add slack variables s_1 and s_2 to convert the inequalities into equalities:

- $32x_1 + x_2 - 12x_3 + s_1 = 10$ (for the second constraint)
- $3x_1 - x_2 - x_3 + s_2 = 0$ (for the third constraint)

Step 2: Set up the objective function and constraints:

The objective function to maximize is:

$$z = 100x_1 + 2x_2 + 5x_3$$

Now we have the constraints:

1. $14x_1 + 2x_2 - 6x_3 + 3x_4 = 7$
2. $32x_1 + x_2 - 12x_3 + s_1 = 10$
3. $3x_1 - x_2 - x_3 + s_2 = 0$
4. $x_1, x_2, x_3, x_4, s_1, s_2 \geq 0$

Step 3: Analyze the problem for possible solutions:

We now analyze if there is an **unbounded solution** or a **no-solution** scenario.

- **Unbounded Solution:** An unbounded solution occurs if the objective function can increase infinitely without violating any constraints. This happens when a variable in the objective function can increase without limit, as long as other variables remain non-negative, which might happen if a constraint is too loose and allows this increase.
- **No Solution:** A problem will have no solution if there is a conflict between the constraints, such as inconsistent inequalities or contradictory restrictions.

Now, let's examine the given options:

Option Analysis:

- **Option A: Solution**
 $x_1 = 100, x_2 = 0, x_3 = 0$: This would be a specific solution, but we need to check if this satisfies all constraints. We cannot make assumptions without checking the constraints thoroughly.
- **Option B: Unbounded Solution:** Given the structure of the inequalities and constraints, it's possible that the objective function might be unbounded if there are no constraints that strictly bound one of the variables (e.g., x_1 or x_3). This could lead to an unbounded solution.
- **Option C: No Solution:** This would occur if the system is inconsistent, but there's no clear indication that this is the case based on the constraints.
- **Option D: Solution**
 $x_1 = 50, x_2 = 70, x_3 = 60$: This is another candidate solution, but we need to check if it satisfies the constraints.

Conclusion:

Option B (Unbounded solution) appears to be correct, especially if you find that the constraints do not prevent some variables from growing infinitely.

Q84 Text Solution:

To calculate $P(A)$, where A represents the event that an even number of buffers are full, we proceed as follows:

Given:

Sample space: $S = \{0, 1, 2, \dots, 32\}$

Probability distribution: $p(i) = (33 - i) / 561$

Event A : Set of even numbers in S : $A = \{0, 2, 4, \dots, 32\}$

Steps:

1. Calculate $P(A)$:

The probability of A is the sum of $p(i)$ for all even i in S :

$$P(A) = \sum p(i) \text{ for } i \in A = \sum (33 - i) / 561 \text{ for } i \in A$$

2. List even numbers in A :

$i \in A = \{0, 2, 4, \dots, 32\}$. There are 17 even numbers in total.

3. Substitute $p(i)$ for each $i \in A$:

$$P(A) = (1 / 561) \sum (33 - i) \text{ for } i \in A$$

4. Calculate the sum:

The sum of $(33 - i)$ for $i \in A$ can be expressed as:

$$\text{Sum} = (33 - 0) + (33 - 2) + (33 - 4) + \dots + (33 - 32)$$

Simplify this:

$$\text{Sum} = 17 \times 33 - (0 + 2 + 4 + \dots + 32)$$

5. Sum of even integers:

The sum of the first n even integers is given by:

$$0 + 2 + 4 + \dots + 32 = (n / 2) \times (\text{largest even number}) = (17 / 2) \times 32 = 272$$

Substituting back:

$$\text{Sum} = 17 \times 33 - 272 = 561 - 272 = 289$$

6. Calculate $P(A)$:

$$P(A) = (1 / 561) \times 289 = 0.515$$

Final Answer: $P(A) = 0.515$

Q85 Text Solution:

Given statement is: $\neg \exists x Q(x)$ is :

This negation will change the quantifier and also it negates the element with the quantifier.

So, it becomes: $\forall x \neg Q(x)$

Because negation of existential is universal quantifier or vice versa.

Q86 Text Solution:

In this question, we have to define which type of numbers are given in the set $A_i = \{-i, \dots, -2, -1, 0, 1, 2, \dots, i\}$

Various type of number with their representation symbol are:

Rational numbers:

A rational number is any real number that can be written as a fraction or in p/q form where p and q are integers. Rational numbers are denoted by Q .

Complex number:

Complex numbers are represented in the form of $a + bi$ where a and b are real numbers and i stands for iota which has only two values either -1 or 1 . Complex numbers are denoted by C .

Real numbers:

Real numbers include both rational and irrational numbers. These are denoted by R .

Integers:

Integers are like whole numbers, but they also include negative numbers. The range is from $-\infty$ to $+\infty$. They are denoted by Z .

So, set $A_i = \{-i, \dots, -2, -1, 0, 1, 2, \dots, i\}$ is the set of all integers which are denoted by Z .

Q87 Text Solution:

$\forall x \forall y (f(x) = f(y) \rightarrow x = y)$, that means if two functions maps same value then input of the functions should be same. This is definition of injective (or one-to-one) function. An injective function or injection or one-to-one function is a function that preserves distinctness: it never maps distinct elements of its domain to the same element of its codomain.

$\forall y \exists x (f(x) = y)$, that means for all y , there is a mapping function from x . This is definition of surjective (or onto) function. A function f from a set X to a set Y is surjective (or onto), or a surjection, if for every element y in the codomain Y of f there is at least one element x in the domain X of f such that $f(x) = y$.

$\forall x f(x) = k$, that means for all x , the output or mapping is only k and never changes. This is



definition of constant function. A constant function is a function whose (output) value is the same for every input value. For example, the function is a constant function because the value of is 4 regardless of the input value.

Therefore, option (D) is correct.

Q88 Text Solution:

The correct option is B.

For a relation to be equivalent, it must be Reflexive(xRx), Symmetric(If xRy then yRx), and Transitive(If xRy and yRz then xRz).

Option 1 is not equivalence as $(1,1)$ is not present and hence violates reflexive property.

Option 2 is equivalence as it is reflexive, symmetric, and transitive.

Option 3 is not equivalence as it fails at reflexive property.(all the diagonal elements are not present i.e. $(1,1),(2,2),(3,3),(4,4)$).

Option 4 is not equivalence as it fails at the reflexive property. (Since $(3,3)$ is not present)

Q89 Text Solution:

The correct option is D.

Equivalence relation: A relation is said to be an equivalence relation if it is a reflexive, symmetric, and transitive relation.

$$1. \{(f,g) | f(x)-g(x)=1 \forall x \in \mathbb{Z}\}$$

It is not reflexive. As $f(x) - f(x) = 0$ it is not 1. It is also not transitive. So, it cannot be an equivalence relation.

$$2. \{(f,g) | f(0) = g(0) \text{ or } f(1) = g(1)\}$$

This relation is not transitive. for e.g. : Let $(f, g), (g, h)$ in R_2 such that $f(0)=g(0)=1$ and $g(1)=h(1)=2$ and $h(0)=4, f(1)=5$. Here f is not related to h . Only we have a relation given between f and g , g and h . but not between f and h . So, it cannot be an equivalence relation.

$$3. \{(f,g) | f(0) = g(1) \text{ and } f(1) = g(0)\}$$

It is not always true $f(0) = f(1)$ for the reflexive case. So, it is not reflexive. Hence, no equivalence relation.

$$4. \{(f, g) | f(x)-g(x)=k \text{ for some } k \in \mathbb{Z}\}$$

It is a reflexive relation, considered constant as 0. It is also symmetric because the difference will be equal to a constant value. It is also transitive. So, it is an equivalence relation.

Q90 Text Solution:

The correct option is B.

2. The set inclusion relation $\subseteq$ is a partial ordering on the power set of a set S

If we consider the set $S = \{a,b\}$. So possibilities with this set are $\{\emptyset, a, b, ab\}$. As, subset relation is reflexive, antisymmetric, and transitive. So, it is partially ordered. Because if a is subset of b , then b is a superset of a .

Q91 Text Solution:

Explanation:

CMOS stores:

- System date & time
- BIOS/UEFI settings

It is a type of non-volatile memory, but not exactly ROM or EPROM.

It is closest in function to ROM because it stores firmware settings.

CMOS is usually categorized as:

- A small amount of non-volatile memory (similar to ROM)

The Final Answer is: ROM

Q92 Text Solution:

The correct option is D.

S	R	Q_{n+1}
0	0	Q_n (No Change)
0	1	0
1	0	1
1	1	x

$S=R=0$ is the normal resting condition of the SR flip-flop. It does not affect the output state of the flop flop. Both Q_1 and $Q_{(t+1)}$ outputs remain in the logic state they were in before this input condition.



Q93 Text Solution:

The correct option is B.

A decoder is a combinational circuit that has 'n' input lines and a maximum of 2^n output lines.

Therefore, Decoder - n lines to 2^n lines.

The multiplexer is a combinational logic circuit designed to switch 1 of several (2^n) input lines to a single common output line. Therefore, Multiplexer - 2^n lines to 1 line.

A de-multiplexer is a device that takes 1 input line and routes it to one of several (2^n) digital output lines. Therefore, De-multiplexer - 1 line to 2^n lines.

Q94 Text Solution:

The correct option is B.

The logic diagram represents a ripple counter.

In a ripple counter, the clock signal is applied to only one flip-flop, and the output of that flip-flop is used as the clock signal for the next flip-flop. This means that the delay in a ripple counter increases as the number of flip-flops increases.

Q95 Text Solution:

The correct option is D.

110101101 can be written as 0001 1010 1101

$$(0001)_2 = 1$$

$$(1010)_2 = (10)_{10} = (A)_{16}$$

$$(1101)_2 = (13)_{10} = (D)_{16}$$

$$\text{Therefore } (110101101)_2 = (1AD)_{16}$$

Q96 Text Solution:

The correct option is A.

$$(-14)_{10} + (-15)_{10} = (-29)_{10}$$

For representing a negative number in binary, 2's complement technique is used.

- Converting 29 to binary - 00011101
- Now taking 1's complement of 00011101 - 11100010
- And adding 1 to 11100010 to make 2's complement - 11100011.

Q97 Text Solution:

The correct option is D.

- Hardware interrupts that can be delayed when a much higher priority interrupt occurs at the same time are called maskable interrupts.
- Unplanned interrupts while executing a program are called exceptions.
- The Source of the interrupt is in phase with the clock is called a synchronous interrupt.

Q98 Text Solution:

The correct option is D.

There are three types of cache mapping:

Direct mapping.

Associative mapping.

Set-Associative mapping.

Segmented - page mapping - A memory management strategy that combines segmentation with paging is called segmented paging. Memory is partitioned into segments of different sizes after being originally segmented into pages of set sizes. A segment table tracks segment location and associated physical addresses, whereas a page table tracks page location and size. The operating system allows a page of the requisite size and breaks it into segments when a program demands memory. So, the option D will be the correct answer.

Q99 Text Solution:

Explanation:

The K-Map is a method of simplifying Boolean algebra expressions.

K-Map can be explained as an array containing a 2^k number of cells, where k is the number of variables in the Boolean expression to be reduced or optimized.



	CD	00 C'D'	01 C'D	11 CD	10 CD'
B'C'	AB	1	1		1
00	A'B'	0	1	3	2
01	A'B	4	5	7	6
11	A B	1	1	X	X
10	A B'	1	1	X	X
		8	9	11	10

The Final Answer is: Option B

Q100 Text Solution:

The correct option is C.

LDA 8000H - transfers the data from memory location 8000H to the Accumulator.

MVI B, 30H - moves the value 30H to Register B.
ADD B - Adds the value of Register B (30H) to the data in Accumulator which stores data from location 8000H. This instruction increments the value as 8000H by 30H.

STA 8001H - stores the content of the Accumulator to memory location 8001H.

Therefore, the above code reads a number from memory at location 8000H and stores it in memory location 8001H after incrementing it by 30H.

NOTE: UGC NET has given the correct answer D but we find a most appropriate answer as option C.



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